Biochar in Stormwater Best Management Practices

Guidebook for Stormwater Practitioners



Version 1

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CENTER FOR WATERSHED PROTECTION

INFINITESOLUTIONS

American Biochar Institute

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ACRONYMS & ABBREVIATIONS

Acronym/Abbreviation	Definition
ABI	American Biochar Institute
BMP(s)	Best Management Practice(s)
CWP	Center for Watershed Protection, Inc.
TMDL	Total Maximum Daily Load
WHC	Water Holding Capacity
PAHs	Polycyclic Aromatic Hydrocarbons
TrOCs	Trace Organic Compounds
SPLP	Synthetic Precipitation Leaching Procedure
CEC	Cation Exchange Capacity
SSA	Specific Surface Area
MIP	Mercury Intrusion Porosimetry
BSM	Bioretention Soil Media
IWS	Internal Water Storage
BEBM	Biochar Enhanced Bioretention Media
BSM	Bioretention Soil Media
ANSI	American National Standard Institute
ASABE	American Society of Agricultural and Biological Engineers
IBI	International Biochar Initiative
VOM	Volatile Organic Matter
PCBs	Polychlorinated Biphenyls
PFAS	Per- and Polyfluoroalkyl Substances
DOM	Dissolved Organic Matter
PSD	Particle Size Distribution
PPE	Personal Protective Equipment
HSG	Hydrologic Soil Group
USDA	U.S. Department of Agriculture
WSS	Web Soil Survey
NRCS	Natural Resources Conservation Service
GR	Green Roof
EGR	Extensive Green Roof

IGR	Intensive Green Roof
DOC	Dissolved Organic Carbon
PAW	Plant Available Water
SOM	Soil Organic Matter
MC	Moisture Content
QC	Quality Control
CMAC	Continuous Monitoring and Adaptive Control
MAP	Modified Atmosphere Packaging

1 GUIDEBOOK PURPOSE

Biochar has been explored as a media component or amendment for stormwater best management practices (BMPs) to enhance stormwater management. Biochar amendments in BMPs may increase pollutant removal, runoff detention and infiltration, and BMP plant survivability. However, biochar characteristics and how it is used in BMPs impact how biochar affects BMP performance. This guidebook provides essential information, based on current biochar research and implementation in stormwater management, to help stormwater designers and agencies successfully incorporate biochar into BMPs for their regional stormwater management goals. The following is included within this guidebook:

- Background on biochar and potential benefits to stormwater management.
- Different biochar materials and target characteristics for specific stormwater management goals, BMP design, and regional eco-sensitivities.
- Recommended testing requirements for biochar producers, suppliers, and installers.
- Design recommendations for incorporating biochar into eight (8) common stormwater BMPs.
- Recommendations for sourcing, delivery, and storage of biochar and biochar amended materials.
- Specification templates for biochar material and handling, and supplemental specifications for incorporating biochar into existing regional BMP designs.

This guidebook is supplemented by the *Biochar in Stormwater Management Technical Basis Report* (LINK), which details current research supporting recommendations provided in this guidebook. To keep this guidebook focused on the design and installation of biochar amended BMPs, detailed documentation, reasoning, and references are provided in the Technical Basis Report. Practitioners interested in the underlying scientific research are encouraged to consult the report.

The *Biochar in BMPs Fact Sheets* (LINK) provide a graphical summary of biochar benefits and design approach for each of the BMPs discussed in this guidebook. These factsheets can be used by designers and agencies to evaluate if and how biochar-amended BMPs may be beneficial based on their regional stormwater management goals, BMP design requirements, and eco-sensitivities. The factsheets serve as an introduction to the use of biochar in stormwater BMPs and can support preliminary site suitability evaluation, development of regional standards for using biochar in BMPs, or communication with stakeholders familiar with stormwater management. However, this guidebook should be reviewed in detail for any detailed design guidance, specifications, or standards involving biochar-amended BMPs.

The *Biochar in Stormwater Guidebook for Stormwater Practitioners* provides a national framework for designing, specifying, and installing biochar into common stormwater BMPs based on current research on the topic, and should be used as a supplement to, not in place of, local and regional BMP design guidance. The guidebook provides guidance for incorporating biochar into general BMP designs and may not exactly match regionally specific BMP designs. Designers should seek agency input when differences exist when directly connected to how biochar is incorporated. By following the recommendations laid out in this guidebook, biochar-amended BMPs should provide improved performance for targeted stormwater management goals, while reducing or eliminating concerns and risks from using biochar which have been identified in research and by stormwater practitioners. It is important to acknowledge that the application of biochar in stormwater BMPs is still an emerging area of research and development. In response to growing demand, this guidebook was created to provide practitioners with best practices informed by the current state of science. American

Biochar Institute (ABI) and the Center for Watershed Protection (CWP) hope to continue reviewing and incorporating updates into future revisions of this document.

1.1. How to Use this Guidebook

The *Biochar in Stormwater BMPs Guidebook for Stormwater Practitioners* is intended to be used by practitioners that work with stormwater BMP design, implementation, and program management. It is recommended that those using the guidebook already understand local/regional stormwater BMP design goals and eco-sensitivities, as well as site-specific needs, objectives, and stormwater regulations. This guidebook is not to be used in place of regional stormwater management regulation or guidance, but to provide opportunities to enhance BMP functions through the addition of biochar into regional stormwater practices.

Section 2 of the guidebook provides a background on how biochar can be beneficial to stormwater BMP performance. Common stormwater pollutants and performance goals are summarized for eight (8) common stormwater BMPs in which biochar can be effectively incorporated. Summaries include how biochar amendments may impact stormwater BMP performance. This section provides context for stormwater practitioners as they begin to explore the potential or expanded use of biochar in BMPs, given their regional stormwater goals.

Section 3 of the guidebook will familiarize stormwater practitioners with guidance for sourcing biochar that will support their stormwater goals. This section begins with a discussion on the variety of biochar materials and approaches for sourcing material. Additionally, this section provides baseline testing recommendations for biochar in stormwater BMPs, as well as additional testing recommendations for targeting common stormwater pollutants, addressing hydrologic stormwater goals, and exercising caution in eco-sensitive regions. The section concludes with a link to specification templates that can be used and adapted by stormwater practitioners for specifying biochar material and quality control.

Section 4 of the guidebook provides stormwater practitioners with BMP-specific guidance. This section provides detailed design recommendations, and information on target pollutants and expected performance from biochar-amended BMPs, biochar application and incorporation techniques, and potential concerns and mitigation to minimize and reduce risk for eight (8) common stormwater BMPs. Each section is written to be a standalone section for design and implementation purposes. The section concludes with a link to specification templates for each BMP. The templates can be used and adapted by stormwater practitioners for specifying biochar application and incorporation into relevant regional BMPs.

Section 5 of the guidebook provides details on biochar sourcing, delivery, and installation. The section reviews typical biochar sourcing, delivery, and installation challenges, and provides guidance and specification wording to mitigate risk to personnel and/or BMP failure.

Section 6 is an assembly of specifications linked throughout the document. Template specifications provide standard language for the incorporation of biochar into BMPs, with cues to the stormwater practitioner as to when regional stormwater management goals and regulations need to be considered.

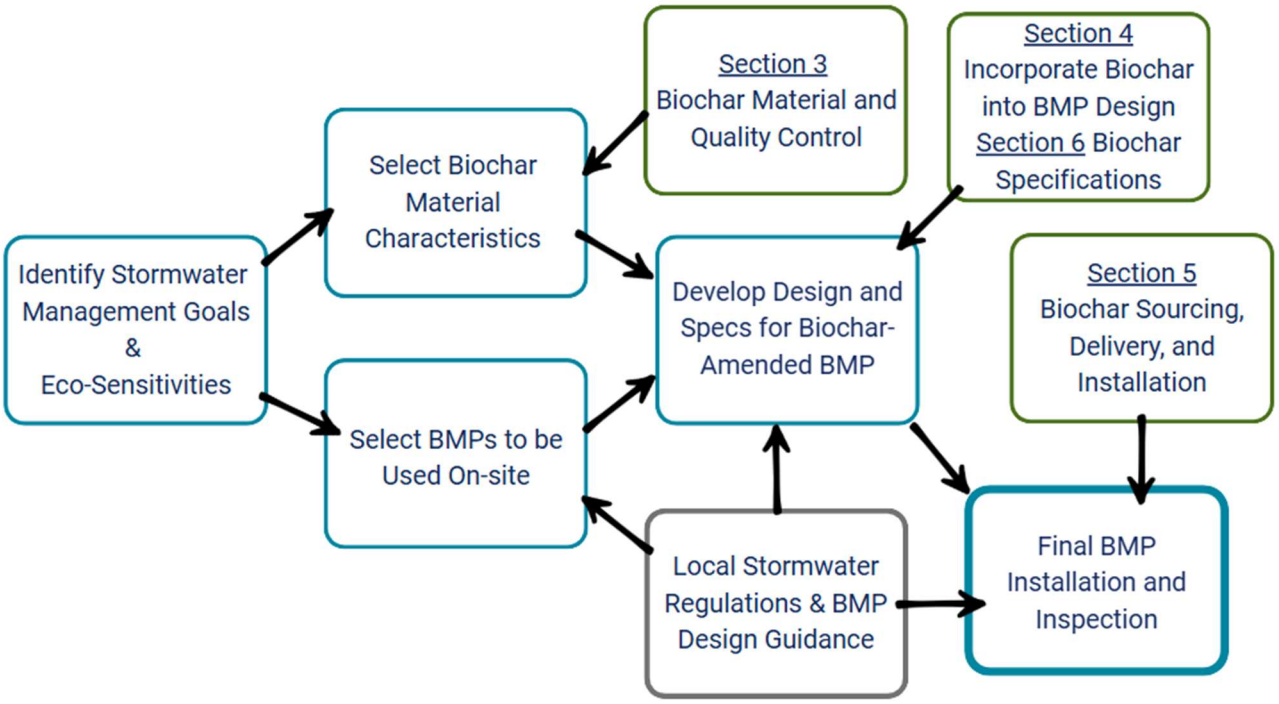


Figure 1. Workflow for using the Biochar in Stormwater BMPs Guidebook for Stormwater Practitioners in conjunction with existing regional stormwater guidance and regulations. Blue boxes indicate BMP design process, green boxes indicate where the guidebook can help inform biochar amendment, and grey boxes indicate incorporation of regional design requirements.

This guidebook provides essential information to stormwater practitioners interested in designing, installing, or evaluating stormwater BMPs that include biochar for improved performance. It also provides essential information that could support the inclusion of biochar in regional stormwater management manuals. However, guidebook users should always exercise best engineering or expert judgements. There are many available biochar and stormwater experts available if guidebook users have further questions. Please contact ABI or CWP with additional questions or for a biochar expert in your region.

2 BIOCHAR BACKGROUND AND BENEFITS

Biochar, defined as a black, carbon-rich material created when biomass undergoes pyrolysis under oxygen-free or oxygen-limited conditions, is currently being explored as a media amendment to enhance stormwater management within best management practices (BMPs). Biomass feedstocks used to produce biochar typically consist of waste materials like crop residues, yard/food/forestry waste, and animal manures. The ultimate performance and usefulness of biochar in stormwater BMPs are significantly influenced by variability in the feedstock, and production conditions including the heating temperature, the processing time, and the management of oxygen in the system. Pyrolysis, a process in which biochar is typically heated to between 450°C and 700°C, is the most direct thermochemical conversion process to produce biochar. Gasification, involving temperatures typically higher than 750C with the presence of some oxygen, is another common approach for producing biochar. As biomass undergoes high heat processing, the carbon in that biomass transitions into polyaromatic hexagonal carbon structures which are highly resistant to decay in the environment, resulting in a material that persists for hundreds to thousands of years. As a physical material, biochar is highly porous, containing macropores that reflect the cellular structure of the initial feedstock and micropores, which are created at higher production temperatures, contain abundant surface area.

Biochar is incorporated into stormwater BMPs to improve infiltration, water holding, and pollutant removal thereby enhancing flow mitigation and pollutant reduction performance. Biochar can be effective for removing heavy metals, nitrogen, phosphorus, bacteria, trace organic compounds (TrOCs), and emerging toxic contaminants (ETC). Abundant research on biochar performance suggests several key benefits that support its use in stormwater BMPs, such that, despite the challenges in developing standardized sourcing and implementation guidance, it should be considered as an amendment to stormwater BMPs in many areas.

One major benefit of biochar amendment is pollutant removal, which is achieved through multiple mechanisms. For heavy metals, removal occurs primarily through adsorption, facilitated by cation exchange or surface complexation with functional groups on the biochar surface. Biochar amendment has been consistently shown to reduce copper and zinc from stormwater. Nitrogen removal is also enhanced, with biochar-amended media tending to outperform sandy media mixes. Ammonium can be removed via adsorption to negative functional groups, while nitrates and nitrites can be reduced through denitrification occurring in the micro-anoxic zones within biochar pores. For biological contaminants, such as Escherichia coli (E. coli) and Enterococci, the increased porosity and surface area of biochar, combined with the hydrophobicity of some biochar, increases the potential for trapping bacteria within the media. High production temperature wood-based biochar, in particular, has shown efficacy in reducing E. coli concentrations. Trace organic compounds (TrOCs), which include various categories of contaminants like pesticides and poly aromatic hydrocarbons (PAHs), are initially removed by adsorption, but biochar also enhances long-term removal by increasing water retention, which provides more time for microbial degradation to occur in pore spaces.

Beyond pollutant removal, biochar offers significant hydrologic benefits, critical for runoff reduction. Biochar amendment increases the water holding capacity (WHC) in soils, particularly in sandy soils such as bioretention soil media (BSM), which is important for detaining or retaining runoff for extended periods, allowing natural removal processes (like denitrification) to occur. This increased WHC is attributed to the porosity of biochar but also its ability to enhance soil agglomeration, leading to increased pore space and pore size diversity. Biochar amended soils also tend to have lower bulk densities, reducing compaction. Furthermore, biochar amendment can increase infiltration properties in fine and medium-grained soils and reduce infiltration rates

in sandy soils, by shifting the particle size distribution (PSD) and improving soil structure via macropores in biochar and soil agglomeration.

Another crucial benefit of biochar is enhanced plant survivability. The majority of biochar research, and indeed the majority of biochar use globally is as a soil amendment to improve plant health and crop yields. In BMPs, plants are essential, supporting evapotranspiration and pollutant removal through uptake, hosting and supporting beneficial microbial communities in root zones, and maintaining infiltration rates via rooting action. Biochar improves soil structure and captures and stores runoff and associated nutrients in pores, improving nutrient uptake and making water available to plants during drought conditions. This leads to increased plant vigor, health, and survivability, which can improve public perception of stormwater BMPs and reduce long-term operation and maintenance costs associated with plant replacement. One way that biochar can be used effectively in stormwater BMPs is as a partial replacement for compost, providing long-lived organic matter that exports only limited amounts of nutrients.

3 BIOCHAR MATERIAL AND QUALITY CONTROL

Biochar is a black, highly porous, carbon-rich material produced when biomass is subject to pyrolysis or gasification. Common biomass for biochar production includes waste materials, such as crop residues, yard/food/forest waste, and animal manures/biosolids. Variability in the pyrolysis or gasification conditions, including heating temperature, process duration, oxygen availability, and feedstock all impact biochar performance and usefulness in stormwater BMPs.

3.1. Biochar Processing

3.1.1. Feedstock

The biomass used to produce biochar is called feedstock. Feedstock can be categorized as woody (e.g., wood, nut shells), herbaceous (e.g., rice and oat hulls, straw, corn stover), or manures/biosolids. Plant-based biochar feedstocks tend to have less potential contamination (i.e. heavy metals, TrOCs, etc.) than sludge or manure-based biochar. Of the plant-based feedstocks, woody biochar typically contains low levels of nutrients for plant growth but offers potential for nutrient capture. Biochar from herbaceous feedstock tends to be highly friable, breaking into fine particles, and posing challenges for flow rates in certain stormwater BMPs. Therefore, it is recommended that high-carbon woody biochar should typically be used in stormwater BMPs due to their low potential for nutrient export and also due to their good flow characteristics. Additional technical experts and research understanding is needed to determine the applicability of high-mineral biochar (i.e. typically produced from non-wood-based feedstocks) in certain stormwater-related applications.

3.1.2. Pyrolysis Temperature

During pyrolysis and gasification, the carbon in biomass transitions progressively towards a polyaromatic hexagonal carbon structure with abundant surface area and micropores as production temperature increases. At the same time, volatile hydrogen and oxygen compounds are removed from the biochar with increasing production temperature. Further, surface functional groups (e.g., carboxyl, phenolic, carbonyl) on the edges of the polyaromatic carbon structures, which provide reactivity and cation exchange capacity, are primarily composed of oxygen and hydrogen and are also volatilized at higher production temperatures. The combined effect of increased production temperature tends to be a decrease in reactivity and sorptivity associated with surface functional groups, but a large increase in surface area, counteracting the loss of functional groups. Over time, slow microbial action on the edges of the polyaromatic carbon structures adds surface functional groups back onto biochar.

For most stormwater applications, pyrolysis temperatures between 400°C and 600°C will stabilize carbon structure, maximize porosity and surface area, maintain a reasonable initial CEC, and provide a structure that can support higher CEC over time as a result of microbial activity. It is recommended that biochar produced from pyrolysis at temperatures less than 400°C not be used in stormwater BMPs due to the potential for organic contaminants, such as PAHs, to be present.

3.2. Classification and Standards

Due to the relatively recent emergence of the biochar industry, only a few standards exist that are applicable in the United States. For biochar analysis, the most applicable standard in the United States is the recently developed ANSI / ASABE / USBI S668: Methods for the Testing and Analysis of Biochar (LINK) approved in late 2025. This analysis standard was developed by a technical committee through the American National Standard Institute (ANSI) process as a standard that could be implemented by commercial laboratories in North America. This standard contains recommended biochar analysis methods for the following parameters relevant to using biochar in stormwater: moisture content, total and organic carbon, ash and volatile matter content, total nitrogen and phosphorus, metals, pH, liming, electrical conductivity, specific surface area, bulk density, and TrOCs.

For material standardization, the International Biochar Initiative (IBI) developed a Biochar Certification Program as a voluntary initiative for standardizing biochar characterization and production; however, this program was terminated in 2024. Among other guidance, the document provides requirements for the analysis of biochar materials, which is ultimately useful for classifying biochar for use. While no longer maintained, this standard still provides useful information for the use of biochar in soil, however, the analysis methods required under this standard are difficult for most laboratories to implement. While this is currently a gap, there are organizations working to develop American standards for biochar in specific uses, including the development of this guidebook for stormwater BMP applications.

3.3. Biochar Characteristics and Performance

Biochar characteristics ultimately determine its effectiveness in stormwater BMPs; however, they vary significantly between different biochar producers, and they can even vary between batches from the same producer. To support the selection of biochar, this guidebook provides recommendations for testing and verifying biochar. The following is a list of characteristics, in addition to feedstock and pyrolysis temperature, that stormwater practitioners should consider when incorporating biochar into stormwater BMPs. Many of these parameters are included in recommended material specifications in Section 6 , Table A, and Table B.

3.3.1. Moisture

Moisture content is a standard biochar analysis parameter and is included in all testing protocols, partially because it is required to calculate other parameters on a dry weight basis. In general, all other biochar analytical parameters should be reported on a dry weight basis.

In the field, moisture affects the handling, storage, and mixing of biochar. If biochar is too moist (>50%) it becomes heavy and can become "cakey", making it more difficult to distribute evenly within soil or a media mix. However, if biochar is too dry (<20%), although unlikely, it can pose self-ignition concerns and can be dusty during mixing and placement.

3.3.2. Total and Organic Carbon

Total carbon is the % of carbon, by mass, within a biochar sample. Carbon in biochar can consist of organic carbon and inorganic carbon, which are the portions of carbon that are contained in combustible organic compounds and the portion of which is contained primarily in carbonate minerals

in ash, respectively. The carbon content of biochar is reflective of both the initial feedstock and the production temperature, increasing at higher production temperatures as oxygen and hydrogen are volatilized. High carbon biochar materials are generally more effective in stormwater BMPs.

3.3.3. H:C_{org} Ratio

The H:C_{org} ratio compares the amount of hydrogen to organic carbon, typically reported as a molar ratio (i.e., moles of H to moles of organic carbon). As noted, with increasing production temperature, hydrogen is volatilized while a portion of the carbon in the original biomass is converted into highly stable polyaromatic forms, resulting in progressively declining H:C_{org} with increasing production temperature. Direct analysis of carbon form is not common, standardized, or affordable, so H:C_{org} ratio is often used as a proxy of biochar chemical form, biochar surface area and porosity, and chemical and biological stability in the environment. H:C_{org} is impacted by both feedstock and pyrolysis temperature. Biochar produced from woody biomass using high-heat pyrolysis has high organic carbon content resulting in a lower H:C_{org} ratio than biochar produced from other feedstocks.

3.3.4. Total Ash

Ash refers to the amount of inorganic residue remaining after a biomass or biochar sample is fully combusted. Ash can contain nutrients, minerals, and sometimes unwanted compounds that may leach from the biochar into stormwater or block surface functional groups. Wood-based biochar has minimal inorganic compounds and tends to inherently have a low ash content.

3.3.5. Volatile Organic Matter

Volatile organic matter (VOM) measures the amount of organic material that did not fully carbonize under pyrolysis and is composed primarily of hydrogen and oxygen compounds. It is strongly correlated with production temperature, with lower temperature biochar containing more VOM and higher temperature biochar containing less VOM. High VOM can be indicative of tars and oil compounds retained within biochar, some of which can contain PAHs and other potential pollutants. Over time, this VOM is slowly degraded by soil microbes, however, in general biochar with low VOM is preferred for use in stormwater BMPs.

3.3.6. Total Nitrogen and Total Phosphorus

Most biochar contains relatively small amounts of nitrogen and phosphorus. Total nitrogen, in particular, is low in nearly all biochar as most nitrogen-containing compounds are volatilized during the pyrolysis process. Phosphorus compounds are more often left behind in ash compounds, however, most feedstock contains only limited phosphorus, resulting in biochar only slightly enriched in phosphorus. Among biochar, manure and biosolids biochar can contain substantial phosphorus and some nitrogen, as both can be abundant in those feedstocks. Both nitrogen and phosphorus are essential plant macronutrients, so they support healthy plant growth, however, in stormwater systems they more often pose a risk of nutrient export.

The availability of nitrogen and phosphorus in biochar depends on the chemical form of each. Some nitrogen in biochar is present as soluble nitrate, nitrite, and ammonium, but the majority is typically contained in organic nitrogen forms which are only slowly available. Phosphorus in biochar is commonly contained in relatively low solubility minerals in the ash fraction. The minerals dissolve very slowly and at low concentrations, such that the overall water quality risk from phosphorus is typically

quite low, except for biochar with higher concentrations of phosphorus like those derived from biosolids and manure.

In general, wood-based biochar, as recommended for stormwater BMPs, tends to have low nutrient fractions. One effective use of wood biochar in stormwater BMPs is as a partial replacement for compost, replacing the organic matter and pollutant removal benefits but without creating the same amount of nutrient pollutant export as compost.

3.3.7. Pollutant Leaching: Nitrogen, Phosphorus, Metals

Pollutant leaching tests measure the amount of nutrients or other pollutants washed from a biochar sample using column tests or batch extract tests. These tests can be valuable when stormwater BMPs are being built in areas with receiving waters that are highly sensitive to certain pollutants, often nitrogen and phosphorus. Typical leaching tests, such as the Synthetic Pollutant Leaching Procedure (SPLP), run deionized water or a weak acid through a small column of media to measure pollutants that leach out of the sample. SPLP analysis can be effective for measuring worst-case leaching potential of biochar, but it was designed to mimic the potential for (mildly acidic) rainfall to leach pollutants at landfills, which is not representative of stormwater which typically has a pH close to neutral. This mildly acidic solution will tend to overestimate phosphorus export due to increased dissolution of mineral phosphate compounds at pH 5.0 compared to a neutral pH.

Most common leaching tests also measure pollutant export in batch tests or in flow through tests using a small amount of water. Most stormwater BMPs treat runoff from large impervious areas, thus treating dozens or hundreds of times more water per year than the volume of the engineered media. As a result, column and batch leaching tests reflect worst case scenario for pollutant export and are primarily representative of the initial operational life of biochar (or any other media). During this time, there can be a significant pulse of pollutants that dissipates rapidly.

It is recommended that pollutant leaching tests for biochar be conducted with modifications that mimic stormwater chemistry and that are done to reflect medium or long-term leaching potential. This can be done by:

- Using leaching water that is either neutral or buffered at neutral pH (preferred) and preferably contains a mixture of other compounds representative of stormwater (i.e., synthetic stormwater).
- Displacing at least 10 pore volumes of leaching water before collecting samples for analysis if using a column testing approach.
- Discarding the decanted water from the first batch test and only completing pollutant analysis on the decanted water from the second batch when using a batch testing approach (e.g., SPLP).
- Screening to reduce fine particles which can disproportionately contribute to turbidity and initial flush of salts, nutrients, and residual compounds.

3.3.8. Metals

It is recommended to test biochar for common metals, including copper (Cu), cadmium (Cd), lead (Pb), nickel (Ni), Selenium (Se), Zinc (Zn), arsenic (As), chromium (Cr), and mercury (Hg). Total

metals testing supports material compliance and quality control, while leach testing is used to evaluate the potential for metals to mobilize under stormwater conditions. Where stormwater goals target specific metals, more stringent acceptance limits and lot-specific verification may be warranted. Clean, untreated wood-based biochar is generally low in metals, but this is dependent on strict exclusion of treated, painted, or demolition wood feedstocks.

3.3.9. pH & Liming

Biochar pH can impact how it interacts with soil/media and stormwater pollutants. Pollutant removal mechanisms in neutral pH (6-7) are dominated by hydrogen bonding, π-π interactions, and ligand exchange whereas removal mechanisms in alkaline environment (pH >8) are dominated by precipitation, electrostatic attraction, and ion exchange. These mechanisms impact which pollutants are removed, as organic pollutants and some metals tend to be removed more effectively at neutral pH, whereas metals and phosphate tend to be removed at higher pH.

Biochar is almost always alkaline, primarily due to the presence of alkaline minerals in the ash fraction. Thus, adding biochar to soil typically increases pH, although often only mildly (i.e., less than 0.5 units) because biochar typically has only limited liming capacity (i.e., pH buffering impact) compared to materials like agricultural lime.

It is recommended that biochar be analyzed for both pH and liming capacity. Generally, biochar with pH less than 10 but liming value less than 10% as $CaCO_3$ is most suitable for stormwater applications, corresponding to common values in wood biochar.

3.3.10. Electrical Conductivity & Salinity

Electrical conductivity (EC) of an aqueous biochar suspension is an indicator of soluble ions (salts) that may be released from the material. Elevated EC can reflect higher concentrations of readily soluble constituents such as potassium, calcium, sodium, chloride, and other ions. Because soluble constituents and ash can be enriched in the fines fraction, screening/gradation control to limit fines can help reduce first-flush export of soluble salts and associated turbidity during commissioning and early operation. While some soluble ions may benefit plant growth in soil applications, in stormwater BMPs excessive EC can contribute to first-flush export of salts and dissolved constituents. Where soils or receiving waters are sensitive to salinity, allowable EC should be limited and confirmed through lot testing; commissioning/initial flush provisions may also be appropriate. Clean wood-based biochar typically exhibits relatively low EC, but values vary by production conditions and handling.

3.3.11. Cation Exchange Capacity (CEC)

Cation exchange capacity (CEC) of biochar reflects its ability to retain and exchange cations including ammonium, magnesium, calcium, and other metals. CEC is particularly important when prioritizing heavy metal removal in stormwater and may also help reduce total nitrogen through retention and the ultimate denitrification of ammonium. Much of the CEC in biochar tends to be pH dependent, with higher CEC at higher pH due to fewer protons occupying otherwise negatively charged surface functional group sorption sites. Biochar produced at lower temperatures tend to have higher CEC than biochar produced at higher temperatures, at least initially. CEC tends to increase as biochar ages and therefore a biochar's initial CEC may not be a good indicator of biochar performance over time.

Analysis of CEC in biochar is feasible, but methods have not been standardized for biochar. Some methods of CEC analysis are known to overestimate CEC in biochar because they mischaracterize soluble cations in ash compounds as CEC. This is a concern for biochar with high ash content not for low ash woody biochar. Further, some CEC methods do not control solution pH and can overestimate the pH-dependent CEC component of biochar which increases solution pH much more than it increases soil pH. In general, CEC analysis of biochar is not always required, but when done, pH-controlled methods with the least potential for ash interference should be used.

3.3.12. Specific Surface Area (SSA)

Specific surface area (SSA) of biochar material represents the total surface area of the material contained in pores. SSA is critical for chemical reactions and pollutant and nutrient sorption. Biochar pollutant removal mechanisms are nearly all dependent on surface interactions and should be considered along with CEC when targeting cation removal. Most of the biochar's surface area is contained in nanopores that are created at high biochar production temperatures due to the re-organization of organic carbon into polyaromatic structures.

Surface area analysis of biochar can be done according to standardized gas isotherm sorption methods; however, SSA analysis is expensive and also difficult to implement accurately. For the majority of stormwater uses, $H:C_{org}$ can be used as a lower cost proxy, with values below 0.4 generally reflecting surface area above 300 m^2/g for woody biochar.

3.3.13. Porosity

Porosity is a measure of the volume, connectivity, and size distribution of pores within biochar. Porosity and SSA are closely related, as surface area exists within biochar pores. However, porosity, measured as pore size and pore size distribution, impacts pollutant access to the biochar surface area, as well as water retention and availability to plants. Further, pore size distribution impacts pollutant removal directly as larger pollutant molecules and pollutants chelated with larger dissolved organic carbon compounds (e.g., Cu^{2+}) cannot physically enter nanopores but are effectively sorbed in larger mesopores and macropores. Biochar with high porosity in diverse pore size ranges is desired for managing stormwater and associated pollutants because of its ability to remove a range of pollutants, retain water, and provide microbial habitat. High temperature woody biochar has among the highest porosity and pore size distribution.

Mercury intrusion porosimetry (MIP) is the most common approach for estimating total porosity and pore size distribution of biochar. However, this approach is expensive, and not currently common analysis practice for most biochar producers. In applications where improved retention of stormwater runoff needs to be quantified or modeled, biochar experts can assist with interpreting MIP and selecting the most appropriate biochar product. However, in most cases, $H:C_{org}$ can be used as a lower cost proxy for porosity, with values below 0.4 reflective of abundant and diverse pores, at least in biochar made from woody feedstock.

3.3.14. Bulk Density

Bulk density is the weight per volume of biochar. Biochar bulk density is dependent on porosity, particle packing, and moisture content. High-porosity biochar tends to have lower bulk density due to the amount of air space within biochar particles, however, when such biochar is wet its bulk density

can still be relatively high. Biochar with lower bulk density can be less expensive to transport but can also be difficult to handle and install due to potential for airborne particles, wind-loss, and uneven field spreading. Refer to biochar specifications for recommendations for working with low bulk density biochar. Unlike most other biochar characteristics, bulk density should typically be reported on an as-received basis rather than on a dry basis.

3.3.15. Toxic Contaminants

Some biochar can contain TrOCs, such as PCBs and PAHs, however, the presence of these is relatively rare when biochar is made from clean feedstocks under proper production conditions.

It is extremely rare for wood-based biochar to contain such toxins, and it is only possible when biochar is produced under low-heat pyrolysis. This guidebook recommends only testing for PAHs when biochar has been produced at low temperatures. Though rare, to mitigate against these potential contaminants, analysis is recommended, and many biochar producers can provide such data.

3.4. Testing and Submittals

Biochar verifications and testing are essential to ensure BMP function. Table A and Table B (in Appendices) summarizes recommended parameters and testing responsibility for biochar before it is incorporated into stormwater BMPs based on target pollutants, and stormwater management goals. Commercial biochar producers have basic testing data available and are typically required to test biochar periodically and each time the feedstock changes. It is also recommended that biochar vendors, including those that mix biochar into bioretention media perform their own verification testing on biochar to ensure their specific product, as well as some parameters are liable to change, namely PSD, during the handling and transportation process. These data will include the biochar characteristics indicated in Table A and Table B. Further, all biochar producers should be able to provide information regarding feedstock, operating conditions, and biochar production temperature.

The on-site installer may also want to test for moisture, bulk density, and particle size distribution of materials received to ensure these properties meet requirements after transport and mixing.

Ultimately, the stormwater practitioner should request and review these data through the submittals process, and such submittals should be built into regional stormwater guidance. A specification template for biochar material and quality control is provided in Section 6 - Specification Templates of this guidebook.

4 BIOCHAR IN BMP DESIGN

This section of the guidebook provides stormwater practitioners with BMP-specific guidance. Detailed design recommendations, information on target pollutants, expected performance from biochar-amended BMPs, biochar application and incorporation techniques, and potential concerns and their mitigation to minimize and reduce risk are included for eight (8) common stormwater BMPs.

- Bioretention with Underdrains
- Bioretention without Underdrains
- High Flow Media Filters
- Vegetated Swales
- Soil Amendment
- Filter Strip & Embankment
- Green Roof
- Dry Pond

Each BMP design subsection is written as a standalone section for design and implementation purposes. Specification templates for each BMP are provided in Section 6 - Specification Templates of this guidance document. Links to the relevant specification templates are provided within each BMP design subsection. The templates can be used and adapted by stormwater practitioners for specifying biochar application and incorporation into relevant regional BMPs.

4.1. Bioretention with Underdrains

Bioretention, also known as bioinfiltration, bioswale, or rain garden; consists of a shallow depression (typically between 3-24 inches) with underlying layers of engineered soil, also known as bioretention soil media (BSM), and stone storage, and underdrain. Underdrains are commonly used in bioretention BMPs areas with low underlying soil infiltration rates.

Runoff is treated in a bioretention by passing vertically through the BSM. Treated runoff is discharged through the underdrain or infiltrates into native soils. Even in areas with low underlying soil infiltration, a significant amount of runoff infiltrates into soils. Bioretention is typically planted with native plants but may also include conventional landscaping or turf.

The material components of bioretention media vary by location. The main component is sand, typically 60% to 95% by volume. The other most common component is organic matter, often compost, but sometimes sawdust or woodchips. Additional media components that target specific pollutants may also be included. Biochar can be added in place of some of the organic components or sand component. However, the total organic component should not exceed regional regulatory thresholds or be greater than 40%, whichever is less.

Gravel storage is common at the bottom of bioretention BMPs, particularly those with underdrains, and varies in depth based on local requirements and BMP design goals. Gravel storage is typically washed gravel and is specified in local design standards. The underdrain in the gravel layer is typically elevated off of the bottom of the gravel layer to encourage infiltration into underlying soils, even in areas with lower native soil infiltration rates.

Bioretention can be designed to increase nitrogen removal by including internal water storage (IWS). IWS uses an upturned elbow or weir at the outlet, creating a saturated zone within the bottom 6 to 12 inches of the stone storage area.

4.1.1. BMP Use

Bioretention is used for the retention and treatment of smaller storm events. Runoff is retained by ponding in the bioretention depression and also in the gravel layer below the underdrain. Pollutant reduction occurs not only via physical filtering and sorption in the bioretention soil media, but also by sedimentation in the ponded area during larger storms. Bioretention can be effective at reducing sediment, particulate nitrogen and phosphorus, fecal indicator organisms, peak flow, runoff volume, heavy metals, and organic contaminants such as polycyclic aromatic hydrocarbons (PAHs). The incorporation of IWS improves bioretention effectiveness at removing dissolved nitrogen components, such as nitrate and nitrite.

Pollutant reduction via sorption (i.e. heavy metals, PAHs, dissolved phosphorus) is very dependent on the composition of the bioretention soil media (BSM). For example, higher compost content increases the cation exchange capacity (CEC) of the bioretention soil media, which increases sorption of cations such as heavy metals. However, most stormwater pollutants exhibit only limited mobility in soils and in groundwater; notable exceptions being nitrate and PFAS.

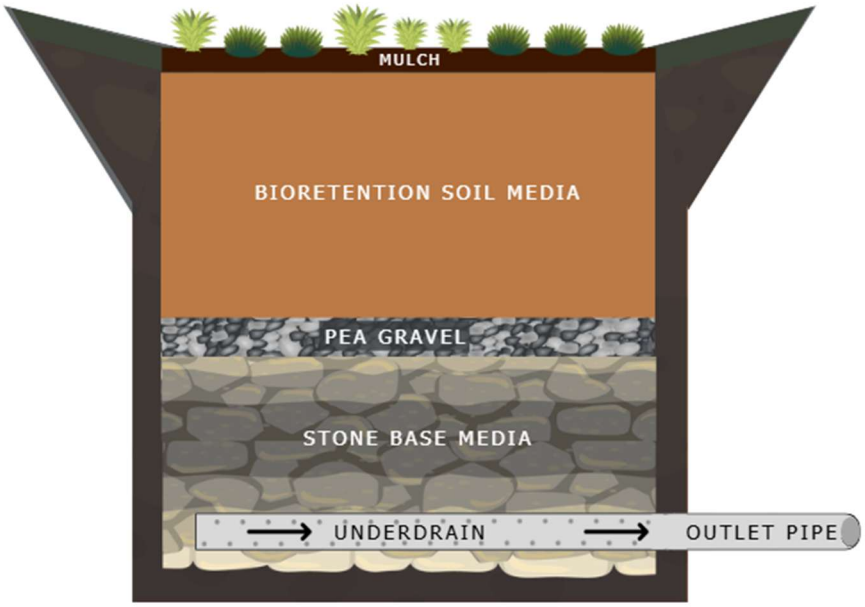


Figure 4.1.1. Graphic of an example bioretention design with an underdrain.

Plant productivity and survivability within BSM is also improved with higher compost fractions within bioretention soil media. It is important to note that higher compost content has also been linked to increased dissolved nutrient leaching from bioretention underdrains.

4.1.2. Benefits of Adding Biochar

4.1.2.1. Verifiable Benefits

Biochar can be added to sandy bioretention media to diversify the particles size distribution and decrease media infiltration rate (K_{SAT}) which slows stormwater runoff as it moves through the media allowing for beneficial treatment and transport processes to occur. Biochar-enhanced bioretention media (BEBM) can increase water retention by directly increasing porosity within bioretention media and by enhancing the formation of soil aggregates. This increased retention can decrease total runoff volume, as the retained water is more likely to undergo evapotranspiration or infiltration. Increased water retention can also improve plant growth and survivability in harsh urban environments or drought conditions. Plant health in bioretention systems is critical for microbial processes to provide nutrient reductions, for plant root growth to help maintain underlying soil infiltration rates, for improved general aesthetics, and reduced maintenance of the BMP.

BEBM can also improve removal of certain stormwater pollutants. Heavy metals, such as copper (Cu), nickel (Ni), and lead (Pb), are consistently removed at higher rates using BEBM. Removal of other heavy metals, including arsenic (As), chromium (Cr), and zinc (Zn), have more mixed results when using high-carbon (i.e. wood-based) biochar amendments. Arsenic and Chromium removal can be complicated due to redox reactions increasing their solubility, especially under anoxic conditions. Whereas Zinc removal is minimally impacted by the inclusion of biochar when compared to other BSM amendments. Removal rates of bacteria, including *E. coli* and *Enterococci*, can be increased using BEBM compared to typical bioretention soil media.

The benefits of biochar in bioretention media are greatest when added to bioretention media with high sand and minimal compost components verse bioretention media with a higher compost fraction. However, because compost leaches nutrients and because it degrades in soil relatively quickly, biochar presents a potential replacement for some of the compost fraction and will provide many of the same plant health and pollutant removal benefits without significant nutrient leaching, all while being effective for a longer period of time.

4.1.2.2. Potential Benefits

Biochar amendments to bioretention mix may increase removal of some trace organic compounds (TrOCs). The best reductions in research have been observed in laboratory trials using biochar with small particle size distributions and stormwater with low dissolved organic matter (DOM), as the DOM competes for sorption sites and interacts with TrOCs in ways that interfere with sorption onto bioretention soil media. Research on using biochar to remove PFAS from soil and from water in laboratory trials, is encouraging, however, these results have not been replicated in the few field trials that have been done to date. This guidebook recommends further investigation to best determine how and what biochar characteristics may help target specific TrOCs.

Biochar amendment in bioretention media reduces nitrates through denitrification in anaerobic pore spaces, but IWS more effectively reduces nitrogen as more water is retained for denitrification. Biochar would have the most overall impact on nitrogen reduction in existing bioretention systems where IWS and associated anaerobic zones were not already present.

4.1.2.3. Limited or No Benefits

Biochar amendment does not directly enhance bioretention ability to remove phosphorus, either in the particulate or dissolved forms. However, it does support healthy vegetation, which increases plant uptake of phosphorus, which studies show to have minimal and potentially unquantifiable reductions.

4.1.3. Biochar Application

There are two recommended ways to incorporate biochar into bioretention media:

- Mixing into bioretention media at vendor site prior to delivery (preferred method), and
- Amending existing bioretention with biochar.

4.1.3.1. Mixing Biochar-Enhanced Bioretention Media

Soil vendors are specially equipped to blend bioretention soil media, therefore, it is recommended BEBM be mixed by vendors with BSM experience. Biochar should be incorporated at a rate of 10% to 20% by volume, replacing either the compost or sand fraction. In general, the total organic component of BSM should not exceed regional regulatory thresholds or be greater than 40%, whichever is lower. If biochar at the recommended application rate is cost-prohibitive, adding less biochar (minimum of 5%) will still provide benefits, but to a lesser extent.

Biochar can be brittle, and extra handling or tumblers for mixing should be avoided. Soil vendors blending biochar into BSM should ideally use less destructive mixing techniques such as paddle mixers or rotary drum mixers. However, when such equipment is not available, or when mixing must be completed on site using bucket loaders, the "bubble wrap" method should be used by mixing biochar first with the compost fractions, then adding the mixture to the sand. Mixing can be achieved by lifting and turning the pile several times with a front-end loader and should be performed on bare ground or an asphalt pad. Mixing should not be performed on grass.

It is recommended that a sample of the BEBM be taken to ensure the particle size distribution (PSD) after mixing is consistent with sand and/or BSM local requirements. Stockpiled BEBM should be stored and maintained as described in Section 5 -Biochar Sourcing, Delivery, and Installation and as outlined in the related Specification Templates in Section 6 .

4.1.3.2. Amending In-Situ Bioretention Media with Biochar

Biochar can be amended into functional, but underperforming bioretention BMPs to enhance performance or plant survivability. Amending in-situ bioretention with underdrains should be treated like amending in-situ soils. Please refer to Section 4.5 - Soil Amendment. Caution should be taken when working around underdrains to ensure equipment does not damage them.

4.1.4. Potential Concerns or Considerations

Stormwater experts typically have two concerns about incorporating biochar into bioretention – pollutant leaching and soil clogging.

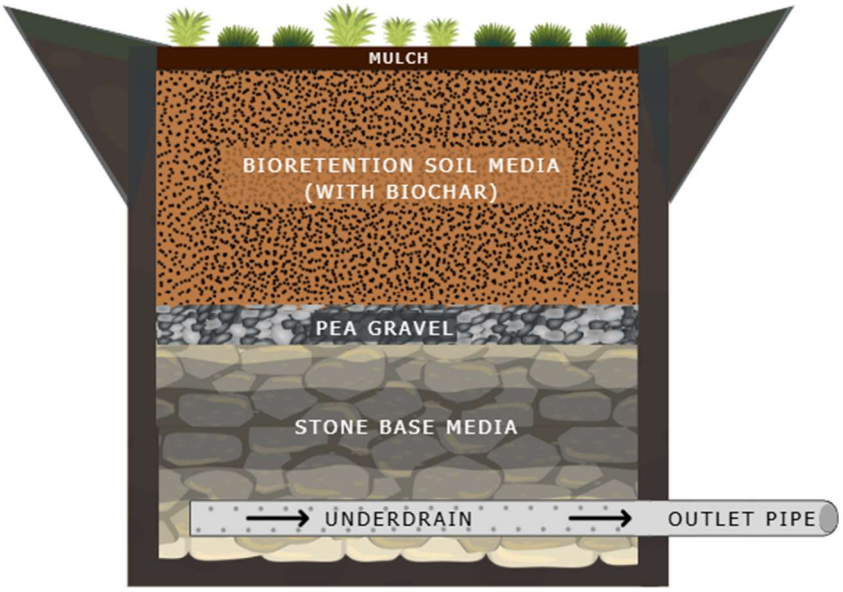


Figure 4.1.2. Graphic of an example bioretention design with an underdrain. The biochar is incorporated into the bioretention soil media (BSM) layer.

4.1.4.1. Nutrient and Contaminant Leaching

In general, biochar is produced under various conditions using a variety of feedstock sources, some of which may have nutrients, heavy metals, salts, or other contaminants within biochar particles. As discussed in detail in the *Biochar in Stormwater BMPs Technical Basis Report* (LINK), woody feedstock inherently has less potential contamination including nitrogen, and biochar becomes cleaner as it is processed under higher-heat conditions. Therefore, it is recommended that wood-based, high production temperature biochar be used in stormwater BMPs. Specifications for the recommended biochar, including testing methods and thresholds, are provided in Section 6 Specification Templates, as well as Table A and Table B in the Appendices. Additionally, contaminants that may impact live plants, such as salts, media should be washed prior to installation or the system can be flushed prior to planting live plants to ensure ash-associated contaminants are not present. When following these specifications, pollutant leaching from biochar is not a major concern.

4.1.4.2. Clogging

Bioretention soil media is typically sandy, which ensures high infiltration rates, but is not conducive to water retention or supporting plant health. Biochar soil amendment can improve media performance, but infiltration rates can be reduced if too fine a biochar is used, or if biochar is aggressively handled while blending BEBM. Biochar PSD should reflect overall BEBM media PSD, falling between 0.5 mm and 3 mm, with the majority falling between 1 and 2 mm. Care should also be taken in handling biochar (i.e. transporting, mixing, and placing), as biochar can easily be broken into smaller particles. It is recommended that the vendor and/or contractor mixing BEBM collect a representative sample for PSD analysis after mixing to ensure fines are not present. Proper techniques and accounting reduce the potential for media clogging.

4.1.5. Specifications for Incorporating Biochar into Bioretention

The following specification templates are provided to guide material selection, storage and handling, installation, and construction sequencing for incorporating biochar into bioretention BMP:

- **Biochar – Material and Quality Control**
- **Biochar-Enhanced Bioretention Media (BEBM)**
- **Biochar Amendment for In-Situ Bioretention**

4.2. Bioretention without Underdrains

Design guidance for bioretention without underdrains can differ by region, but the goals are like those of bioretention with underdrains. There are two main scenarios when an underdrain is not needed for bioretention:

- Where in-situ soils have a high permeability, where a standard bioretention profile with bioretention media can effectively provide pollutant removal and retention.
- Where connected impervious surface and ponding depth are minimized in soils with medium to low permeability. A BMP designed under this scenario can also be referred to as bio-basin or rain garden.

Bioretention without underdrains consists of a shallow depression with underlying layers of engineered soil, also known as bioretention soil media (BSM). Runoff enters a bioretention and, during larger storms, ponds above the media surface while slowly infiltrating into the media. Treated runoff leaves the bioretention by infiltrating into native soils. The depression is typically planted with native plants that tolerate temporary inundation.

Bioretention soil media (BSM) recipes vary by location. The main ingredient is sand, typically making up 60% to 95% by volume. If in-situ soils are categorized and field confirmed as hydrologic soil group A, some regions will allow for in-situ soils or topsoil to replace or be incorporated into the sand component of the bioretention media. The other most common component is organic matter, often compost, but sometimes sawdust, woodchips, or coconut coir. Additional BSM components that target specific pollutants may also be included.

A gravel storage is sometimes included in the design of bioretention without underdrains to provide additional storage for enhanced infiltration. The storage layer will vary in depth based on local requirements and BMP design goals. While this layer is typically washed gravel, additional specifications or requirements are found in local design standards.

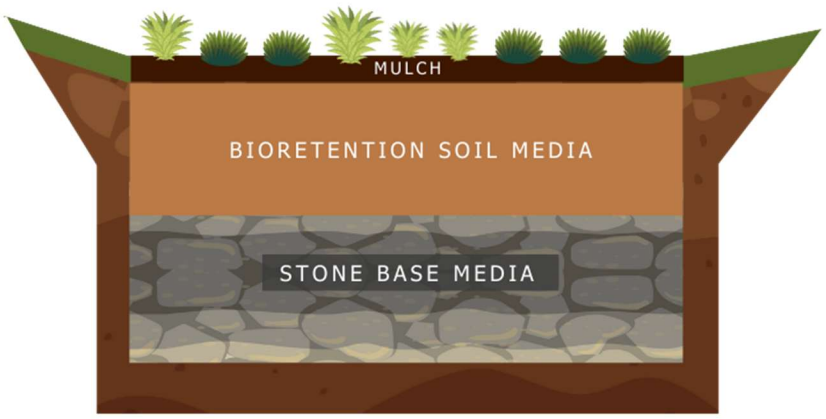


Figure 4.2.1. Graphic showing an example design for bioretention without an underdrain.

4.2.1. BMP Use

Bioretention is used for the retention and treatment of smaller storm events. Runoff is retained by ponding in the bioretention depression and in the gravel storage layer, when present. Pollutant reduction occurs not only via physical filtering and sorption in the bioretention soil media but also occurs via sedimentation during larger storms when surface ponding occurs. Bioretention can be effective at reducing sediment, particulate nitrogen and phosphorus, fecal indicator organisms, peak flow, runoff volume, heavy metals, and organic contaminants such as polycyclic aromatic hydrocarbons (PAHs).

Pollutant reduction via sorption (i.e. heavy metals, PAHs, dissolved phosphorus) is very dependent on the composition of the BSM. For example, higher compost content increases the cation exchange capacity (CEC) of the BSM, which increases sorption of cations such as heavy metals. However, most stormwater pollutants exhibit only limited mobility in soils and in groundwater; notable exceptions being nitrate and PFAS.

Plant productivity and survivability within BSM is also improved with higher compost fractions within bioretention media. It is important to note that higher compost content has also been linked to increased dissolved nutrient leaching from bioretention with underdrains. While this concern is mitigated in bioretention without underdrains, it is still a concern for nitrate, which is generally highly mobile in soils and groundwater. Dissolved phosphorus tends to exhibit lower mobility in soils and groundwater, except in certain cases with high water tables and high organic matter content.

4.2.2. Benefits of Adding Biochar

4.2.2.1. Verifiable Benefits

Biochar can be added to sandy bioretention media to diversify the particles size distribution and decrease media infiltration rate (K_{SAT}) which slows stormwater runoff as it moves through the media allowing for beneficial treatment and transport processes to occur. Biochar can also be used to shift the PSD of the media from silty to sandy, increasing K_{SAT} to increase media infiltration potential. Biochar enhanced bioretention media (BEBM) can increase water retention by directly increasing porosity within bioretention media and by enhancing the formation of soil aggregates. This increased retention can decrease total runoff volume, as he retained water is more likely to undergo evapotranspiration or infiltration. Increased water retention can also improve plant growth and survivability in harsh urban environments or drought conditions. Plant health in bioretention systems is critical for microbial processes to provide nutrient reductions, for plant root growth to help maintain underlying soil infiltration rates, for improved general aesthetics, and reduced maintenance of the BMP.

BEBM can also help improve pollutant retention in bioretention without underdrains, however, most common stormwater pollutants are not highly mobile in soils or in groundwater, so these benefits are limited. Among pollutants, nitrate is a key exception, being highly mobile in soils and in groundwater. In areas where nitrate pollution is a concern, biochar can be used as an effective alternative to compost, providing many of the same benefits without being a source of nitrate pollution.

4.2.2.2. Potential Benefits

Biochar amendments to bioretention mix may increase removal of some trace organic compounds (TrOCs). The best reductions in research have been observed in laboratory trials using biochar with small particle size distributions and stormwater with low dissolved organic matter (DOM), as the DOM competes for sorption sites and interacts with TrOCs in ways that interfere with sorption onto bioretention soil media. Research on using biochar to remove PFAS from soil and from water in laboratory trials, is encouraging, however, these results have not been replicated in the few field trials that have been done to date. This guidebook recommends further investigation to best determine how and what biochar characteristics may help target specific TrOCs.

Biochar may also reduce total nitrate leaching to groundwater. Numerous studies have documented reductions in nitrate leaching to groundwater following biochar as a soil amendment in agricultural soils. This is thought to be related to the retention of nitrates in pore water and also to retention of ammonium which is commonly converted into nitrate via nitrification. While these mechanisms are relevant in agricultural systems, bioretention BMPs treat much larger amounts of water, so these mechanisms may be relatively small in the context of stormwater.

4.2.2.3. Limited or No Benefits

Biochar amendment does not directly enhance bioretention ability to remove phosphorus, either in particulate or dissolved forms. However, it does support healthy vegetation, which increases plant uptake of phosphorus, which studies show to have minimal and potentially unquantifiable reductions.

4.2.3. Biochar Application

There are two recommended ways to incorporate biochar into bioretention media:

- Mixing into bioretention media at vendor site prior to delivery (preferred method), and
- Amending existing bioretention with biochar.

4.2.3.1. Mixing Biochar-Enhanced Bioretention Media (BEBM)

Soil vendors are specially equipped to blend bioretention soil media, therefore, it is recommended BEBM be mixed by vendors with BSM experience. Biochar should be incorporated at a rate of 10% to 20% by volume, replacing either the compost or sand fraction. In general, the total organic component of BSM should not exceed regional regulatory thresholds or be greater than 40%, whichever is lower. If biochar at the recommended application rate is cost-prohibitive, adding less biochar (minimum of 5%) will still provide benefits, but to a lesser extent.

Biochar can be brittle, and extra handling or tumblers for mixing should be avoided. Soil vendors blending biochar into BSM should ideally use less destructive mixing techniques such as paddle mixers or rotary drum mixers. However, when such equipment is not available, or when mixing must be completed on site using bucket loaders, the "bubble wrap" method should be used by mixing biochar first with the compost fractions, then adding the mixture to the sand. Mixing can be achieved by lifting and turning the pile several times with a front-end loader and should be performed on bare ground or an asphalt pad. Mixing should not be performed on grass.

It is recommended that a sample of the BEBM be taken to ensure the particle size distribution (PSD) after mixing is consistent with sand and/or BSM local requirements. Stockpiled BEBM should be stored and maintained as described as described in Section 5 -Biochar Sourcing, Delivery, and Installation and as outlined in the related Specification Templates in Section 6

4.2.3.2. Amending In-Situ Bioretention Media with Biochar

Biochar can be amended into functional, but underperforming bioretention BMPs to enhance performance or plant survivability. Amending in-situ bioretention without underdrains should be treated like amending in-situ soils. Please refer to Section 4.5 - Soil Amendment.

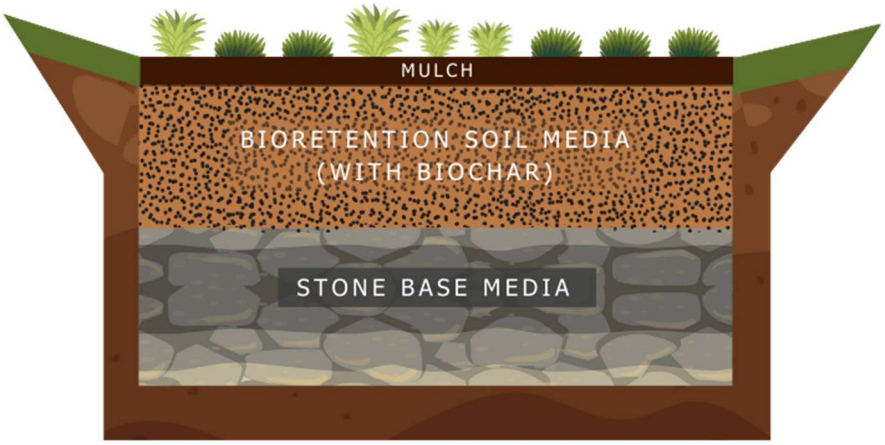


Figure 4.2.2. Graphic of a bioretention without an underdrain where the biochar amendment is incorporated into the bioretention soil media (BSM) layer.

4.2.4. Potential Concerns or Considerations

Stormwater experts typically have two concerns about incorporating biochar into bioretention – nutrient/contaminant leaching and clogging.

4.2.4.1. Nutrient and Contaminant Leaching

Biochar is produced under a variety of conditions using a variety of feedstock sources, some of which may have nutrients, heavy metals, salts, or other contaminants within biochar particles. As discussed in detail in the *Biochar in Stormwater BMPs Technical Basis Report* (LINK), woody feedstock inherently has less potential contamination including nitrogen, and biochar becomes cleaner as it is processed under higher-heat conditions. Therefore, it is recommended that wood-based, high production temperature biochar be used in stormwater BMPs. Specifications for the recommended biochar, including testing methods and thresholds, are provided in Section 6 Specification Templates, as well as Table A and Table B in the Appendices. Additionally, for contaminants that may impact live plants, such as salts, media should be washed prior to installation or the system can be flushed prior to planting live plants to ensure ash-associated

contaminants are not present. When following these specifications, pollutant leaching from biochar is not a major concern.

4.2.4.2. Clogging

Bioretention soil media is sandy, which ensures high infiltration rates. Sand is not conducive to water retention or supporting plant health. Biochar amendment can improve media performance, but infiltration rates can be reduced if too fine biochar is used, or if biochar is aggressively handled while blending BEBM. Biochar PSD should reflect overall BEBM media PSD, falling between 0.5 mm and 3 mm, with the majority falling between 1 and 2 mm. Care should also be taken in handling biochar (i.e. transporting, mixing, and placing), as biochar can easily be broken into smaller particles. It is recommended that the vendor and/or contractor mixing BEBM collect a representative sample for PSD analysis after mixing to ensure fines are not present. Proper techniques and accounting reduce the potential for media clogging.

4.2.5. Specifications for Incorporating Biochar into Bioretention

The following specifications templates are provided to guide material selection, storage and handling, installation, and construction sequence for incorporating biochar into bioretention:

- **Biochar – Material and Quality Control**
- **Biochar – Enhanced Bioretention Media**
- **Biochar Amendment for In-Situ Bioretention**

4.3. High Flow Media Filters

High flow media filters are BMPs designed to rapidly filter stormwater runoff through a specialized media to remove pollutants. They are most used when space is limited or infiltration isn't an option. High flow media filter material is typically proprietary and can incorporate many different media components, including biochar, to target specific pollutants, while allowing for high hydraulic loadings.

High flow media filters can appear similar to bioretention systems, with a depression to allow for ponding water, but they can also use different designs such as horizontal flow. Stormwater runoff is rapidly routed through the media and discharged through an underdrain or manifold to a collection system or surface water. High flow media systems are sometimes but not always vegetated.



Figure 4.3.1. Examples of proprietary high flow media filters

4.3.1. BMP Use

High flow media filters are able to treat larger volumes of stormwater runoff from highly impervious areas where space for green infrastructure BMPs is limited. Runoff is routed to the high flow media filters, either by surface flow or by conveyance, and is typically retained temporarily by ponding on top of, or adjacent to the high flow media. Pollutant reduction occurs via physical filtration, adsorption, and limited biological processes. High flow media filters effectively remove solids, nutrients bound to particulates, hydrocarbons, dissolved metals, and larger pathogens (i.e. cryptosporidium and giardia). High flow media filters are not effective at removing dissolved nutrients, unless specifically targeted in the media composition (e.g., iron filings and granular ferric oxide target dissolved phosphorus).

4.3.2. Benefits of Adding Biochar

4.3.2.1. Verifiable Benefits

Biochar amendment to high flow media can provide many of the same pollutant removal benefits of other components, such as granular activated carbon, but at a lower cost. Specifically, biochar can consistently remove heavy metals, such as copper (Cu), nickel (Ni), and lead (Pb). In some studies, biochar is more effective than granular activated carbon, which is typically not designed for heavy metal removal. Biochar can also effectively replace the hydrocarbon removal benefits of activated carbon.

4.3.2.2. Potential Benefits

Biochar amendments to high flow media may increase the removal of some trace organic compounds (TrOCs). The best reductions are seen using biochar with small particle size distributions (PSD) and stormwater with low dissolved organic matter (DOM), as the DOM competes for sorption sites. These issues also pertain to granular activated carbon, reducing its effectiveness in many stormwater applications. This guidebook recommends further investigation to best determine how and what biochar characteristics may help target specific TrOCs.

Biochar amendment in high flow media may reduce nitrates through denitrification in anaerobic pore spaces, which can improve high flow media performance when removing dissolved nutrients. However, given that microbially mediated denitrification is a slow process relative to high media flow rates, this benefit is likely only relevant for that portion of runoff retained between storm events.

4.3.2.3. Limited or No Benefits

Biochar amendment does not directly enhance high flow media ability to remove phosphorus.

4.3.3. Biochar Application

Biochar can be incorporated into a high flow media at a rate of 5% to 10% at the manufacturer's discretion, or it can be used to replace all or a portion of the granular activated carbon component. Maintaining hydraulic loading capacity is critical to high flow media filter function and design. Therefore, only biochar screened to remove material sized less than 1 mm is recommended. Care should be taken when incorporating biochar into media recipes, as biochar can be brittle and easily broken into smaller pieces. It is recommended that a high flow media distributor work directly with stormwater and biochar practitioners to develop both the correct application, as well as incorporation process to optimize biochar benefits in high flow media filters.

4.3.4. Potential Concerns or Considerations

The greatest concern for incorporating biochar into high flow media filters is that the biochar may reduce hydraulic loading. High flow media is coarse, which ensures high infiltration rates, but is not effective for water retention or supporting plant health. Incorporation of biochar can improve high flow media performance related to both these aspects, but infiltration rates can be reduced if too fine of a biochar is used. Biochar particle size distribution (PSD) is critical to ensure hydraulic loading is maintained and should predominantly fall within 1 mm and 3 mm. Care should also be taken in handling biochar (i.e. transporting, mixing, and placing), as biochar can easily be broken into smaller

particles. It is recommended that the biochar vendor screen the biochar to ensure fines are not present. Further, the high flow media manufacturer should wash the biochar prior to mixing to reduce fines created during transport. Furthermore, the overall PSD of the final mixed high flow media should be verified to ensure biochar was not crushed during mixing. Proper techniques and accounting reduce the potential for high flow media filter clogging.

4.3.5. Specifications for Incorporating Biochar into High Flow Media Filters

The following specifications templates are provided to guide material selection, storage and handling, installation, and construction sequence for incorporating biochar into high flow media filters:

- **Biochar – Material and Quality Control**

4.4. Vegetated Swales

Vegetated swales (a.k.a. swales, ditches) are a green alternative to grey stormwater conveyance such as concrete channels or pipes. Vegetated swales are used to convey stormwater runoff, often seen adjacent to a roadway. Vegetated swales also provide some retention and water quality benefits. It is important to note that some swale designs may include a modified media layer along the bottom of the swale to improve stormwater management and treatment performance. When using such media at the base of a Vegetated Swale they can function similarly to bioretention without an underdrain. Please refer to Section 4.2 - Bioretention without Underdrains of this guidance if considering incorporating a modified media layer into the swale rather that amendment only.

Vegetated swales are trapezoidal, triangular, or parabolic channels that are fully vegetated, often with grass. The type of grass influences the vegetation depth, as stiffer grass species maintain vertical stands better than leafy grass species. Vegetated swales are designed to maintain non-erosive flows and typically have a slope between 0.5% and 2%. If longitudinal slope is steeper, check dams or other flow interruption structures can be used throughout the swale. When flow depths are shallow during small storm events, vegetated swales provide some water quality benefits, including infiltration, water retention, and macrofiltration by grasses. During larger storms, vegetated swales typically provide limited water quality benefits and operate primarily as conveyance.

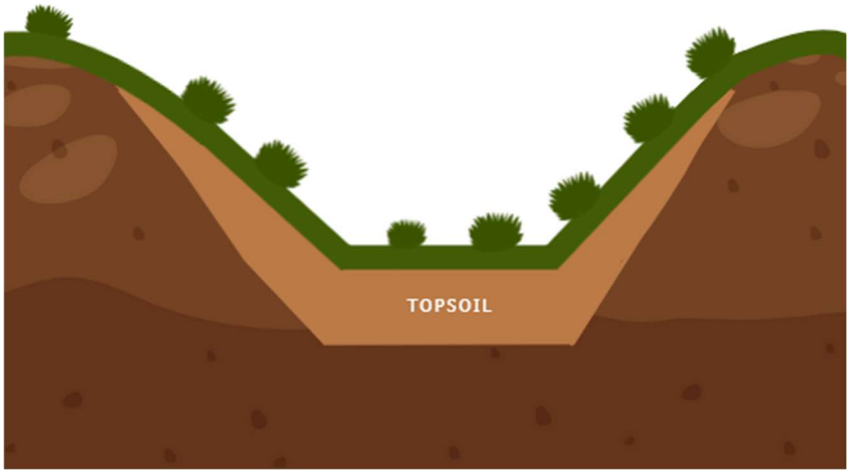


Figure 4.4.1 Graphic showing an example design for vegetated swale.

Biochar amendments to vegetated swales can improve BMP performance and resilience over time by increasing infiltration and water retention, providing opportunity for enhanced denitrification, reducing compaction, and supporting plant health.

4.4.1. BMP Use

Vegetated swales are used predominantly as a green alternative to pipes or concrete channels for stormwater conveyance. They are a cost-effective option for channelizing and directing stormwater

linearly, as they can receive runoff via overland flow throughout the length of the swale. When designed appropriately, a vegetated swale will manage and treat stormwater runoff. Vegetated swales designed to maintain shallow flow depths during water quality storms and/or 1- or 2-year recurrence intervals will also provide water quality benefits. Regional design guidance typically specifies required water quality flow depths based on dominant vegetation species (usually a maximum of 6" to 12" deep). Water quality benefits occur when vegetation maintains a vertical stand during periods of flow to maintain channel roughness, which in turn slows runoff through the swale, ultimately increasing opportunities for infiltration, water retention, and macrofiltration through grasses. If stormflow through the swale is too deep or vegetation gets too long, grasses can fall over and decrease roughness (i.e., Mannings n), effectively creating a flow surface much closer to that of concrete with minimal water quality benefits.

4.4.2. Benefits of Adding Biochar

4.4.2.1. Verifiable Benefits

Although vegetated swales are primarily designed for conveyance, they also provide water quality benefits during small storm events. Biochar amendments to vegetated swales targeting water quality are like those benefits seen with bioretention without underdrains when using in-situ soils. Biochar amendment can increase infiltration by increasing the porosity of amended soils directly due to the high porosity of biochar, but also indirectly by supporting the development of soil aggregates. This increased porosity, helps increase water retention in amended soils which increases infiltration into underlying soils and improves plant survivability in harsh urban environments and during drought. Increased water retention also provides an opportunity for reduction in nitrates through denitrification in anaerobic pore spaces. Runoff retention ultimately decreases runoff volume, and the associated discharge of pollutants to surface waters, as the retained water is more likely to be evapotranspired by plants or infiltrate into underlying soil.

Biochar amendment to vegetated swales may further reduce certain stormwater pollutants. Heavy metals, such as copper (Cu), nickel (Ni), and lead (Pb) are consistently removed at higher rates using biochar amended media mixes. Removal of other heavy metals, including arsenic (As), chromium (Cr), and zinc (Zn), are not impacted by biochar amendments.

4.4.2.2. Potential Benefits

Biochar amendments to vegetated swales may increase removal of some TrOCs. The best reductions are seen using biochar with small particle size distributions and stormwater with low dissolved organic matter (DOM), as the DOM competes for sorption sites. This guidebook recommends further investigation to best determine how and what biochar characteristics may help target specific TrOCs.

Biochar amendment to vegetated swales may reduce pathogens and their associated indicators through hydrophobic attractions. However, research is unclear as to how extended retention and wet periods will impact pathogen reproduction within the system.

4.4.2.3. Limited or No Benefits

The impact of biochar amendment to water quality through vegetated swales is difficult to measure. Pollutants such as hydrocarbons or metals coming in contact with biochar-amended soils

are likely to sorb to biochar, but there is limited research investigating this for vegetated swale applications. Biochar can also improve the health of grasses in swales, which could improve coarse pollutant filtering during smaller storms. However, any such benefits would likely be negligible during larger storms.

4.4.3. Biochar Application

Biochar can be applied to vegetated swales during construction or as a retrofit technique to improve performance of existing swales. The underlying goals of biochar amendment to vegetated swales should be to improve plant health and to improve the soil texture to encourage infiltration into underlying soils. Both goals can be achieved by using biochar as a soil amendment to modify soil texture towards sandy loam, which has the most desirable soil characteristics relative to vegetated swale performance, including moderate infiltration rate and moderate to high water holding capacity (WHC) (Figure 4.4.2).

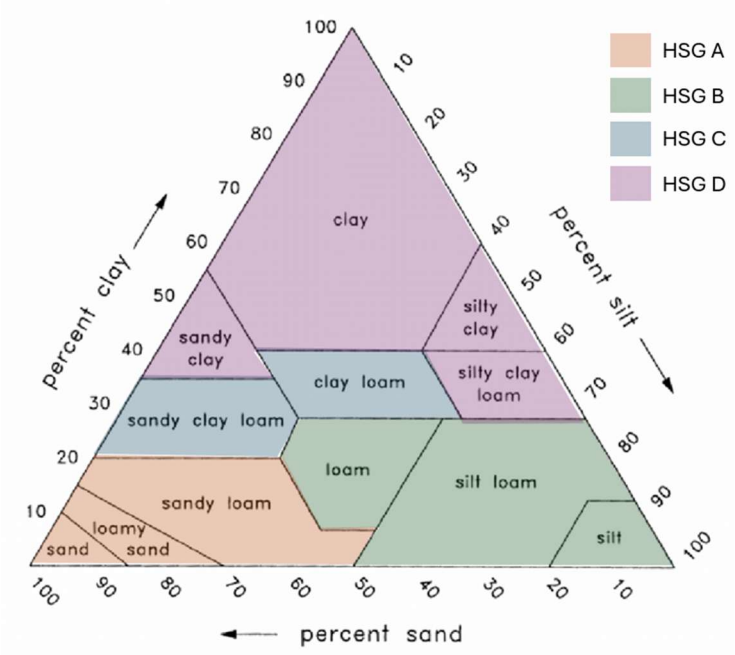


Figure 4.4.2. Soil texture and hydraulic soil group based on particles size distribution

4.4.3.1. Determine In-Situ Soil Texture

The first step in determining the best biochar application rate is to characterize in-situ soil conditions at the depth of the proposed modification. Vegetated swale soils are likely disturbed and not accurately represented on local or regional soil surveys. Therefore, field testing of identified biochar application areas is preferable to soil surveys. Field and lab soil tests may include:

- Particle Size Distribution (PSD) – Sand, silt, and clay composition (recommended).
- Organic Matter Content (% SOM) – Affects soil structure and moisture retention (optional).
- Bulk Density (g/cm³) – Indicates compaction and aeration (required).

- Saturated Hydraulic Conductivity (Ksat, in/hr) – Measures water movement through the soil in field (optional, good for quantifying improvements).
- Infiltration Rate (in/hr) – Determines surface water absorption capacity in field (optional, good for quantifying improvements).

4.4.3.2. Determine Biochar Application Rate

Biochar application rates for amending vegetated swales are typically between 5% and 10% by volume. Use the USDA NRCS Web Soil Survey (WSS) to determine Hydrologic Soil Group for the BMP. Sites with Class A, B and C soils are most suitable for biochar soil amendments, as biochar can be easily worked into these soils. HSG D soils can benefit from certain biochar applications, namely biochar material with low porosity to prevent enhanced retention, but there may be more suitable amendments for achieving similar results (i.e. sand, bioretention media). It is recommended that stormwater practitioners interested in improving stormwater performance within in-situ HSG D soils consult a local soil expert.

Biochar should not be applied at a greater rate than 10% by volume to prevent potential erosion due to concentrated flow through the BMP. If biochar at the recommended application rate is cost-prohibitive, adding less biochar will still benefit the vegetated swale, but to a lesser extent.

4.4.3.3. Application Method

For amending biochar to existing soils, those soils within the vegetated swale should be tilled to a depth of 12 inches using a power harrow tiller or similar vertical mixing/tilling machine (no horizontal tilling). The harrow tiller allows uniform mixing with depth without inverting moist substructure soils and creating hardpan or glazed horizontal layers. Biochar can be evenly added to the surface of the vegetated swale and the power harrow tiller can be used to mix the biochar and in-situ soils. The disturbed area should be properly seeded with an appropriate permanent seed mix and stabilized with matting and check dam structures until vegetation is established.

For incorporating biochar into imported soils, it should be blended with other media components at a rate between 5% and 10% by volume, either replacing a portion of the sand fraction or the organic fraction (often compost). In the case of vegetated swales which are not primarily designed to infiltrate runoff, on-site mixing of biochar with other media components can be used effectively. This approach may increase the portion of fine biochar particles due to breakage during mixing, but this will have limited effect on the performance of vegetated swales.

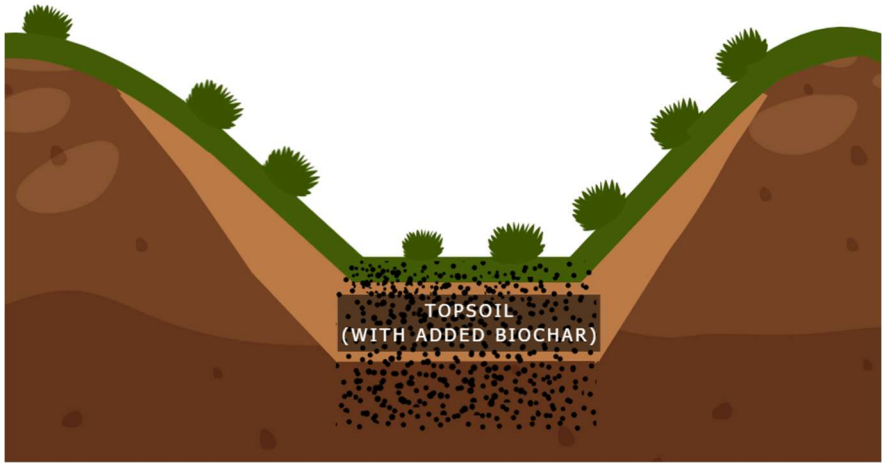


Figure 4.4.3 Graphic of a vegetated swale where the biochar amendment is incorporated into in-situ soil at the bottom of the vegetated swale.

4.4.4. Potential Concerns or Considerations

Major concerns from stormwater experts and installation crews when incorporating biochar into vegetated swales are the challenges associated with moving and placing the biochar material. Biochar is light and can be easily scattered by the wind if not sufficiently moistened. Care should be taken by installers to ensure proper personal protective equipment (PPE) is used during biochar handling. On the other hand, like most media, biochar can become clumpy and difficult to apply uniformly if too wet. Striking the best moisture conditions for safety and ease of installation will come with experience as contractors work with biochar more frequently. To balance these concerns, a target moisture content of 25%-35% is recommended. Due to biochar's ability to conduct current, in-field testing using a probe will require calibration but ultimately may help contractors maintain biochar material workability.

A biochar-amended soil will have a decreased bulk density and be susceptible to erosion in areas of concentrated flow, such as the base of the vegetated swale. Care should be taken to properly protect amended soils using erosion matting and check dams or wattles to prevent soil erosion and allow for vegetation establishment.

4.4.5. Specifications for Incorporating Biochar into Vegetated Swales

The following specifications templates are provided to guide material selection, storage and handling, and installation of biochar into vegetated swales:

- **Biochar Material and Quality Control**
- **Biochar Material Handling**
- **Biochar Material Stockpiling**
- **Biochar Amendment for Vegetated Swales**

4.5. Soil Amendment

Soil amendments enhance existing soils with respect to stormwater management by reducing compaction, increasing infiltration and water retention, immobilizing existing contaminants, and better supporting healthy plant growth. They are especially useful for compacted, modified, polluted, and/or degraded urban soils. Amendments vary from organic inputs (i.e., compost, mulch) to buffers (i.e., lime, gypsum) to specialized media (i.e., iron filings, zeolite), and are typically chosen based on soil need, site-specific stormwater management goal, cost, and availability.

Biochar is another promising soil amendment for stormwater BMPs. It provides similar benefits to organic amendments in improving infiltration and water retention and supporting plant growth with much less concern over nutrient leaching. It can also help immobilize pollutants in urban soil.

4.5.1. BMP Use

Soil amendments as a stormwater BMPs are typically used to enhance desirable stormwater hydrologic and potentially associated water quality – load reduction – goals. Soil amendment is applied to grassed or vegetated areas expected to receive sheet flow runoff, typically within urban areas or adjacent to impervious areas, however, they are not designed to receive substantial hydrologic loading rates. Soil amendments improve hydrologic performance through modification of soil texture, support of microbial and plant communities, and soil structure improvements (i.e., aggregation), all of which help mimic the natural (or predevelopment) hydrologic cycle.

4.5.2. Benefits of Adding Biochar

4.5.2.1. Verifiable Benefits

The benefits of using biochar as a soil amendment are similar to those benefits seen with filter strips and embankments and dry pond retrofits where stormwater function depends on in-situ soil structure. Biochar amendment is used to move a soil texture closer to a sandy loam which is the preferred soil texture for soil amendment BMPs due to its infiltration and retention capacity. This can be done both in overly sandy soils and also overly fine-textured soils. Biochar amendment to sandy soils can decrease infiltration rate, depending on particle size, but will typically increase water retention by increasing soil porosity. Increased water retention from biochar amendment also supports plant growth and survivability in harsh urban environments or during drought conditions. Certain biochar material can be used to amend clayey soils to increase infiltration and decrease bulk density which supports runoff volume reduction, increases root penetration, and supports plant health. However, biochar material with low porosity should be used for these applications to prevent enhanced water retention, as clay soils inherently retain water. Ultimately, there may be more suitable amendments for achieving similar results (i.e. sand, bioretention media). It is recommended that stormwater practitioners interested in improving stormwater performance within in-situ HSG D soils consult a local soil expert.

4.5.2.2. Potential Benefits

Amending soil with biochar will provide some improvements to water quality in stormwater via increased soil health, including soil aggregation and improved infiltration, ultimately reducing site

runoff. Additionally, the use of biochar in place of compost can limit the potential for nitrate export from soil amendment BMPs while still providing many of the same benefits.

Regions across the United States account for pollutant reductions differently, and stormwater practitioners should discern how best to account for water quality improvements through biochar amendments.

4.5.3. Biochar Application

Soils are good candidates for biochar amendment when they exhibit signs of soil compaction or clogging, evident by bare soil patches and standing water and when they receive either direct rainfall or sheet flow run-on. Overly sandy soils can be good candidates to support healthy vegetation. The general goal of biochar amendment should be to reduce compaction and to move the soil texture closer to a sandy loam, which has the most desirable soil characteristics for stormwater management, including moderate infiltration rate and moderate to high water holding capacity (Figure 4.5.1). Shifting in-situ soils to a sandy loam may not be practical, depending on existing soil texture, but any shift in texture towards a loam will improve soil stormwater management performance.

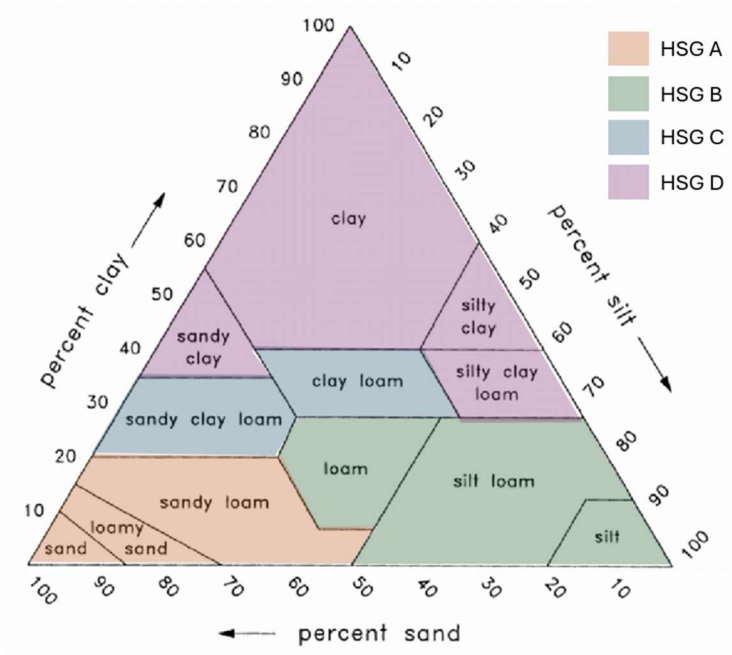


Figure 4.5.1. Soil texture and hydraulic soil group based on particles size distribution

4.5.3.1. Application Rate

The recommended biochar application rates for soil amendment are between 5% and 20% by volume. Use the USDA NRCS Web Soil Survey (WSS) to determine Hydrologic Soil Group (HSG) for the BMP. Sites with USDA Hydraulic Soil Group (HSG) Class A, B and C soils are suitable for biochar soil amendments. However, biochar soil amendments are most compatible with dry, sandy, well-drained soils, and/or soils with low nutrient content that have been heavily compacted – which is common in urban areas. Loamy-sand, sandy-loam, and sandy clay loam can be amended with 5% to 20% biochar by volume. HSG D soils can benefit from certain biochar applications, namely biochar material with low porosity to prevent enhanced retention, but there

may be more suitable amendments for achieving similar results (i.e. sand, bioretention media). It is recommended that stormwater practitioners interested in improving stormwater performance within in-situ HSG D soils consult a local soil expert.

The following application rates are generally recommended:

- HSG Class A soils: 5-10% biochar by volume to improve water retention in sandy soils.
- HSG Class B or C soils: 5-20% biochar by volume to improve infiltration in finer soils.

Biochar should not be applied at rates greater than 20% by volume. If biochar at the recommended application rate is cost-prohibitive, adding less biochar (minimum of 5%) will still benefit the existing soil conditions, but to a lesser extent.

4.5.3.2. Application Method

Application depth depends on the type of vegetation that the soil will be supporting. For turf applications, an amendment depth of 6 to 12 inches is sufficient. For perennial and hardwood plantings, an amendment depth of 12 to 24 inches is recommended to support deeper root penetration which is critical for healthy plants.

Amended areas should be cleared, subsoiled, tilled, amended, and stabilized within the same working day. Therefore, the size of the biochar soil amendment area should be carefully considered to ensure the work can be completed in a single working day. If the area is too large to complete in a single day, it can be broken into smaller areas to complete the amendment over the course of multiple days.

Existing vegetation should be removed, including turf, perennials, and overburden. Tillage for amendments should work outside of existing tree root zones. Subsoil – fracture the compacted soil found at depths 14 inches or deeper – to avoid adversely disturbing the topsoil. A Yeomans or similar plow is recommended for subsoiling the treatment area to the desired depth. Then a rotary power harrow or other similar equipment is used to till the freshly subsoiled soil to a depth of 12 inches. Biochar should then be applied and tilled into the soil in three passes, with approximately 1/3 of the total desired biochar amendment applied during each pass. Once all the required biochar has been thoroughly tilled into the in-situ soils, any tracks should be raked out, and any high areas should be leveled to minimize potential for concentrated runoff or ponding water. The disturbed area should then be properly stabilized and seeded with an appropriate permanent seed mix.

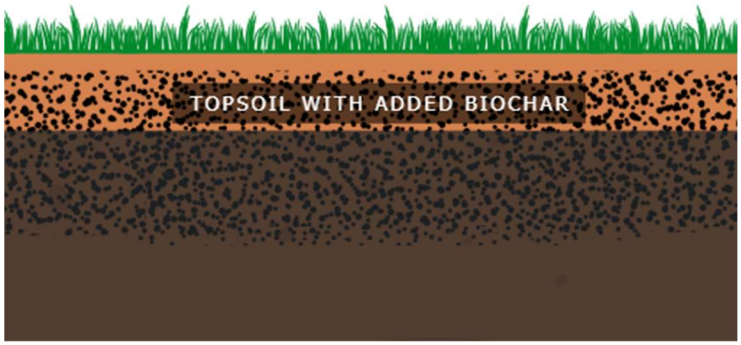


Figure 4.5.2 Graphic of soil amendment where the biochar amendment is incorporated into in-situ soil.

4.5.4. Potential Concerns or Considerations

Major concerns from stormwater experts and installation crews when incorporating biochar into in-situ soils are the challenges associated with moving, placing, and incorporating biochar. Biochar is much lighter than other soil amendments and is easily scattered by the wind if not sufficiently moistened. Installers should be careful to ensure proper personal protective equipment (PPE) is used while biochar handling. It can also be very labor intensive to properly spread the biochar prior to tilling in and methods to avoid damaging the particle size. Like most media, biochar can become clumpy and difficult to apply uniformly if too wet. Striking the best moisture conditions for safety and ease of installation will come with experience as contractors work with biochar more frequently. To balance these concerns, a target moisture content of 25%-35% is recommended. Due to biochar ability to conduct current, in-field testing using a probe will require calibration, but ultimately may help contractors maintain biochar material workability

4.5.5. Specifications for Soil Amendment using Biochar

The following specifications templates are provided to guide material selection, storage and handling, installation, and construction sequence for using biochar as a soil amendment:

- **Biochar Material and Quality Control**
- **Biochar Material Handling**
- **Biochar Material Stockpiling**
- **Biochar as In-Situ Soil Amendment**

4.6. Filter Strip & Embankment

A filter strip is a strip of grass typically installed to intercept sheet flow runoff to provide an initial level of stormwater treatment, usually prior to stormwater entering another BMP or before flowing into a conveyance. Perhaps the most prevalent use of filter strips is grassed embankments along roadways, that discharge into vegetated swales. Filter strips are laterally flat with a minor longitudinal slope (10% max) to maintain sheet flow and nonerosive flows. Runoff routed through filter strips will undergo macrofiltration (through grasses) and some infiltration.

Biochar amendments to filter strips can improve BMP performance and resilience over time by increasing soil storage and infiltration, reducing compaction, and supporting plant health.

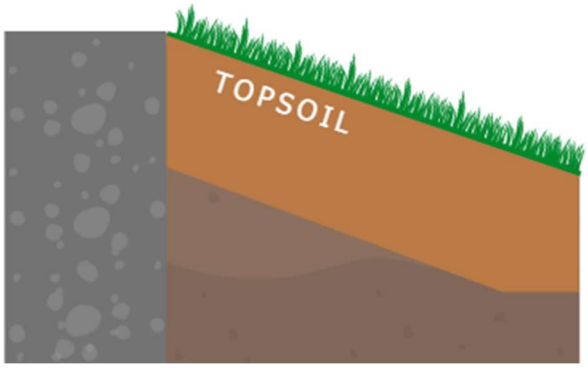


Figure 4.6.1 Graphic showing an example design for filter strip.

4.6.1. BMP Use

Filter strips are used predominantly for pre-treatment and provide stormwater management benefits including macrofiltration and infiltration. Ideally, stormwater runoff is routed evenly across a filter strip as sheet flow to avoid channelization of flows, however, in practice channelization is quite common. Runoff flows through the filter strip in the direction of the slope and is slowed down via vegetation. By slowing the runoff, the filter strip can increase infiltration into surface soils and macrofiltration through vegetation stands. Regionally, filter strips are often attributed to some water quality benefits, but often as pre-treatment and not a stand-alone BMP.

4.6.2. Benefits of Adding Biochar

4.6.2.1. Verifiable Benefits

The benefits of using biochar as an amendment to filter strips are like those benefits seen with soil amendments and dry pond retrofits where stormwater function depends on in-situ soil structure. Biochar amendment is used to support healthy vegetation, reduce compaction, and to move soil texture closer to a sandy loam which is the preferred soil texture for filter strips. Biochar amendment to compacted or clayey soils increases infiltration which supports runoff volume reduction, and it also decreases bulk density, further supporting infiltration, as well as increased root penetration and plant health. However, low porosity biochar should be used in these soils to prevent enhanced water retention.

Biochar amendment to sandy soils, common on roadside embankments can decrease infiltration rate slightly, but increase water retention by increasing porosity within soils. This increased water retention supports plant growth and survivability in harsh environments or drought conditions, which commonly exist on roadside embankments. Because vegetation in filter strips is critical to reduce erosion, a small reduction in infiltration rate associated with biochar addition is outweighed by plant health benefits of adding biochar to sandy soil.

4.6.2.2. Potential Benefits

Biochar amendments to filter strips will provide some improvements to water quality in stormwater via increased soil health, including soil aggregation and improved infiltration, ultimately reducing site runoff. Additionally, the use of biochar in place of compost can limit the potential for nitrate export from soil amendment BMPs while still providing many of the same benefits. Regions across the United States account for pollutant reductions differently, and stormwater practitioners should discern how best to account for water quality improvements in biochar-amended filter strips.

4.6.3. Biochar Application

Biochar can be used in filter strips either during initial construction or as a retrofit after construction. The general goal of using biochar in filter strips should be to improve vegetation health, reduce compaction, and to increase infiltration into underlying soils. Both can be achieved by using biochar to move the soil texture closer to a sandy loam, which has the most desirable soil characteristics for stormwater management. (Figure 4.6.2). Shifting in-situ soils to a sandy loam may not be practical, depending on existing soil texture, but any shift in texture towards a loam will improve soil stormwater management performance.

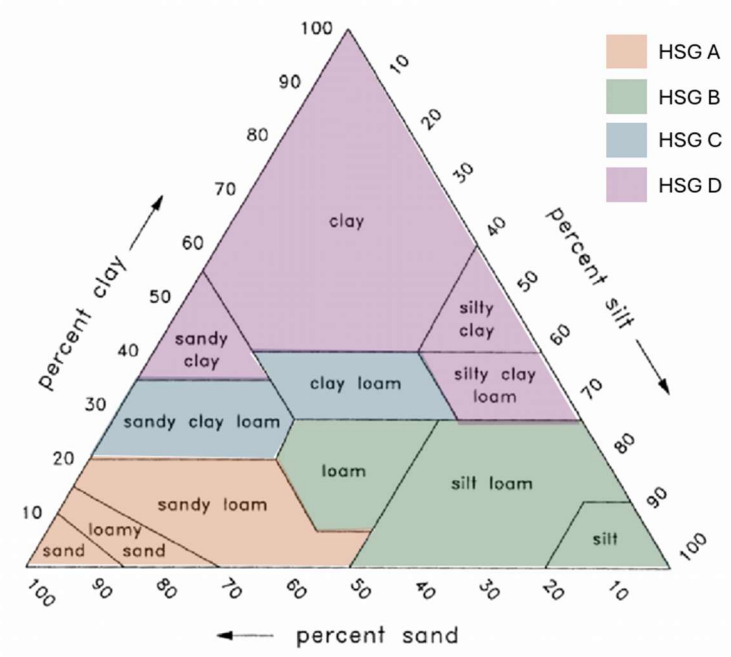


Figure 4.6.2. Soil texture and hydraulic soil group based on particles size distribution

4.6.3.1. Application Rate

Biochar amendments to filter strips and embankments are most compatible with compacted and/or soils with low nutrient content. Such soils are common in the roadside environment, particularly in embankments. Use the USDA NRCS Web Soil Survey (WSS) to determine Hydrologic Soil Group (HSG) for the BMP. Sites with USDA Hydraulic Soil Group (HSG) Class A, B and C soils are suitable for amended filter strips and embankments. Sand, loamy-sand, sandy-loam, and sandy clay loam can be amended with 5% to 20% biochar by volume. HSG D soils can benefit from certain biochar applications, namely biochar material with low porosity to prevent enhanced retention, but there may be more suitable, lower cost amendments for achieving similar results (i.e. sand, bioretention media). It is recommended that stormwater practitioners interested in improving stormwater performance within in-situ HSG D soils consult a local soil expert.

The following application rates are generally recommended:

- HSG Class A soils: 5-10% biochar by volume to improve water retention in sandy soils.
- HSG B or C soils: 5-20% biochar by volume to improve infiltration in finer soils.

Biochar should not be applied at rates greater than 20% by volume. If biochar at the recommended application rate is cost-prohibitive, adding less biochar (minimum of 5%) will still benefit the filter strip and embankment, but to a lesser extent.

4.6.3.2. Application Method

The recommended application depth depends on the type of vegetation that the soil will be supporting. For most applications, where grasses are the predominant vegetation, an amendment depth of 6 to 12 inches is sufficient. For areas of native plantings, an amendment depth of 6 to 24 inches is recommended to support deeper root penetration which is critical for healthy plants. Vegetation species and height should follow regional design guidance, and no changes need to be made to account for the inclusion of biochar in the soil.

Amended areas should be cleared, subsoiled, tilled, amended, and stabilized within the same working day. Therefore, the size of the biochar soil amendment area should be carefully considered to ensure the work can be completed in a single working day. If the area is too large to complete in a single day, it can be broken into smaller areas to complete the amendment over the course of multiple days.

Existing vegetation and overburden should be removed. The treatment area should then be subsoiled – fracturing the compacted soil found at depths 14 inches or deeper – to avoid adversely disturbing the topsoil. A Yeomans or similar plow is recommended for subsoiling the treatment area to the desired depth. Then a rotary power harrow or other equipment should be used to till the freshly subsoiled soil to a depth of 12 inches. Biochar should then be applied and tilled into the soil in three passes. During each pass 1/3 of the total desired biochar amendment should be added. Once all the required biochar has been thoroughly tilled into the in-situ soils, any tracks should be raked out, and any high areas should be leveled out to minimize potential for concentrated runoff or ponding water. The disturbed area should be properly stabilized and seeded with an appropriate permanent seed mix according to regional guidance.

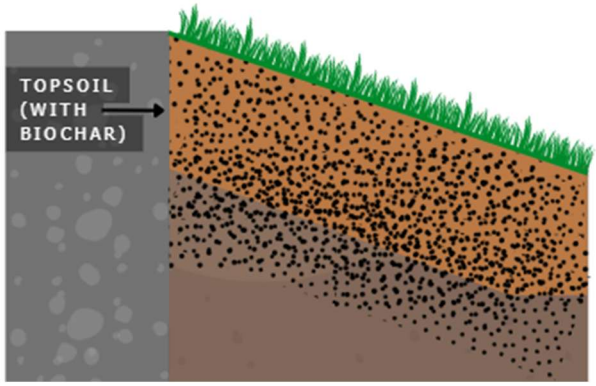


Figure 4.6.3 Graphic of biochar amendment to filter strip where the biochar amendment is incorporated into in-situ soil.

4.6.4. Potential Concerns or Considerations

The most significant concerns among stormwater experts and installation crews about incorporating biochar into in-situ soils are those challenges associated with moving, placing, and incorporating biochar. Biochar is much lighter than other soil amendments and is easily scattered by the wind if not sufficiently moistened. Installers should be careful to ensure proper personal protective equipment (PPE) is used while handling biochar. It can also be very labor intensive to properly spread the biochar prior to tilling in and methods to avoid damaging the particle size. Like most media, biochar can become clumpy and difficult to apply uniformly if too wet. Striking the best moisture conditions for safety and ease of installation will come with experience as contractors work with biochar more often. To balance these concerns, a target moisture content of 25%-35% is recommended and is already the preferred moisture content range among many biochar producers. Due to biochar's ability to conduct current, in-field testing using a probe will require calibration, but ultimately may help contractors maintain biochar material workability.

4.6.5. Specifications for Incorporating Biochar into Filter Strips and Embankments

The following specifications templates are provided to guide material selection, storage and handling, installation, and installation of biochar into filter strips and embankments:

- **Biochar Material and Quality Control**
- **Biochar Material Handling**
- **Biochar Material Stockpiling**
- **Biochar Amendments to Filter Strips and Embankments**

4.7. Green Roof

A green roof (GR) is a living roof system installed atop a building structure featuring vegetation planted in an engineered growth medium, or substrate, above drainage layers. Green roofs serve as a nature-based solution to mitigate adverse environmental effects resulting from urbanization, such as increasing impervious surfaces like rooftops, which lead to hydromodification – alteration of the natural flow of water through a landscape – and water pollution.

The green roof system is layered to perform specific functions. A typical substrate (growing medium) is engineered as a blend, commonly mixing light-weight aggregate materials (such as heat-expanded shale or clay) with an organic component (such as compost). The typical layers include a root-resistant membrane, a drainage layer, a filter layer, the substrate, and the vegetation layer.

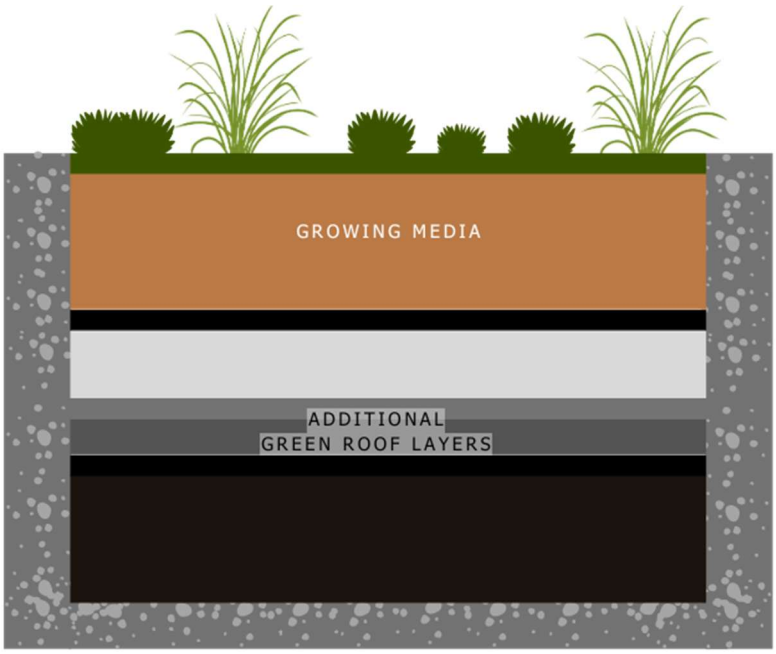


Figure 4.7.1 Graphic showing an example design for green roof.

Green roofs are classified based on the depth of the substrate layer and the resulting weight.

- Extensive Green Roofs (EGRs) feature shallow substrate layers and typically use drought-tolerant vegetation. These systems are widely popular due to their suitability for retrofitting and low structural weight.
- Intensive Green Roofs (IGRs) use deeper substrate layers, which allow for greater plant diversity and typically achieve higher overall water retention. Due to their greater mass, they can require structural design modifications to support the added weight.

4.7.1. BMP Use

A green roof functions as a stormwater BMP primarily through its capacity for runoff retention. The vegetative and substrate layers absorb, retain, and slowly release precipitation, easing the pressure on municipal drainage systems during storm events. The green roof substrate, acting as an artificial soil, must perform the role of retaining water, supplying nutrients, and providing physical support to plants while remaining lightweight, chemically stable, aeratable, and able to drain freely. The proportion of incoming rainfall leaving a green roof as runoff during storm events can vary, with retention averaging 35 to 38% in successfully vegetated treatments, demonstrating significant precipitation retention compared to bare surfaces. In addition to slowing down runoff, green roofs can reduce the total amount of runoff volume via evapotranspiration, particularly when storm events occur during the growing season.

Green roofs provide essential ecosystem services, including stormwater management. While retention is a major benefit, green roofs can also act as a source of pollution via nutrient leaching, particularly phosphorus. Dissolved organic carbon (DOC) concentrations in runoff can also be high in standard green roof systems. The primary challenge in substrate design is balancing the provision of moisture and nutrients essential for plant health and uptake with the need to minimize pollutant export to local waterways.

4.7.2. Benefits of Adding Biochar

Biochar, derived from the pyrolysis of biomass like wood, wood chips, or sawmill waste, is a carbon-rich, stable product that could be widely used as a substrate additive for green roofs due to its light weight, high porosity, and strong water retention capacity.

4.7.2.1. Verifiable Benefits

Biochar amendment to green roof media is predominantly hydrologic, which in turn, improves the media's ability to support vegetation. Biochar-amended green roof media has a higher water holding capacity (WHC) and plant available water (PAW) due to the added porosity of the biochar. When rainwater is retained in the soil media pore structure, it is either used by plants or evaporated, which reduces overall runoff volume from the otherwise impervious roof surface. Green roof plantings are more apt to grow and survive in a harsh urban roof environment when the media is able to retain available water for a longer period of time. When added in appropriate amounts and blends, biochar can be used without negatively impacting plant health and potentially improving it.

Biochar amendments to green roof media may also reduce the bulk density of the media due to its loose and porous structure, depending on the components that are being replaced. Even when fully saturated, the biochar-amended media is lighter than most conventional media, allowing for either an allowable increase in media depth (and rainwater storage) or structurally safe green roof applications for a wider variety of structures.

4.7.2.2. Potential Benefits

Known to leach common stormwater pollutants, green roofs are not typically installed for water quality benefits. However, green roofs using biochar-amended media leach less dissolved organic

carbon and nutrients in its effluent than conventional media, resulting in an overall lower negative impact on water quality.

4.7.3. Biochar Application

To maximize the benefits of biochar in green roofs, the form, application rate, and placement should be optimized. The use of larger biochar particles is recommended to mitigate the risk of biochar washing out of the coarse media that is commonly used as green roof substrate. Intermediate biochar application rates of around 5% to 10% by volume are optimal. The content of biochar in green roof substrate mixtures should not exceed 10% by volume, as negative effects on microbial activity and plant growth have been observed. Because finer particles (< 1 mm) can wash out of media, they should be avoided.

The common method to combine the biochar into the green roof substrate is to thoroughly mix the biochar into the substrate, ideally prior to application. Because the media in many green roof systems is proprietary and built into the design of modular systems, the biochar should ideally be included in proprietary designs so that it can be more widely deployed.



Figure 4.7.2 Graphic of biochar amendment to green roof where the biochar amendment is incorporated into the growing media.

4.7.4. Potential Concerns or Considerations

Before incorporating wood-based biochar into green roof projects, several concerns must be acknowledged.

4.7.4.1. Risk of Biochar Discharge

Fine grained wood-based biochar is highly susceptible to loss due to wind and can also be discharged through coarse media during storms. Therefore, the use of larger biochar particles is recommended in green roofs highly susceptible to these forces to ensure long-term effectiveness.

4.7.4.2. Alkalinity and Plant Health

The liming effect of biochar can increase the substrate pH to an alkaline level, as well as impact the pH of the effluent and overall plant health. In coarse textured soils like green roof blends, the effect of liming is enhanced, so biochar with less than 5% $CaCO_3$ equivalence is preferred.

4.7.4.3. Potential for Clogging

Fine biochar particle sizes can reduce the air-filled porosity and infiltration rate of the green roof media, potentially leading to waterlogging or clogging during or after heavy rainfall events. Therefore, the use of larger biochar particles is recommended to reduce the likelihood of this occurring during the green roof lifetime.

4.7.5. Specifications for Incorporating Biochar into a Green Roof

Green roof substrate is often proprietary. Therefore, mixing specifications are not included within this guidance document. Specifications for biochar material should reference the following:

- **Biochar Material and Quality Control**
- **Biochar Material Handling**
- **Biochar Material Stockpiling**

4.8. Dry Pond

In the past, dry ponds, also known as dry detention basins, dry basins, dry ponds, were installed for stormwater volume management. However, as stormwater management goals have evolved, dry ponds were found to have minimal, and sometimes negative, impacts on water quality. Despite rarely being used as stormwater management for new or redevelopment, many previously installed dry ponds exist throughout the landscape. Therefore, there are many opportunities to improve dry pond performance and reduce maintenance hassle (i.e. unwanted shallow ponding) by amending these existing BMPs.

Dry ponds are earthen basins that are used to temporarily hold and slowly release stormwater runoff during rain events. Unlike wet ponds, also common across landscapes, dry ponds do not have a permanent ponding area, are dry between rain events and vegetated with typical non-wetland plants. Dry ponds provide runoff detention, and the slow release of the detained runoff mitigates storm peak flow while promoting infiltration into in-situ soils. Dry ponds typically consist of a grassed basin, an outlet structure to control release from the pond, and a spillway.

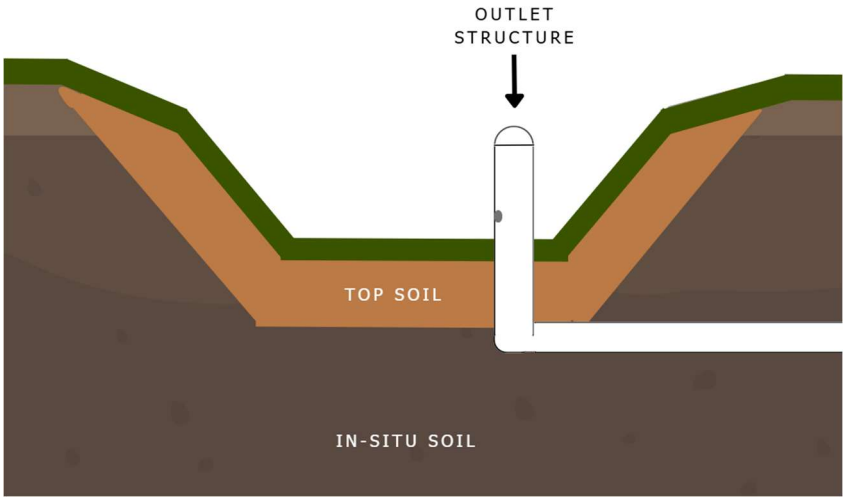


Figure 4.8.1 Graphic showing an example design for dry pond.

Biochar amendments to dry ponds can improve BMP performance and resilience over time by increasing soil storage and infiltration, reducing compaction, and supporting plant health in between rainfall events.

4.8.1. BMP Use

Dry ponds are used predominantly for detention and, if in-situ soils support it, infiltration. Stormwater runoff is routed to dry ponds typically via grey conveyance. Dry ponds typically provide some level of controlled release, which may be a simple outlet pipe, a multi-orifice control structure, or a real time control valve to provide extended ponding (up to 72 hours). During the ponding period, stormwater infiltrates into in-situ soils and may achieve pollutant removal through runoff (i.e. load) reduction. However, as dry ponds drain, sediment and associated pollutants can be resuspended and incorporated into discharge, or re-suspended during the next storm, so dry ponds, without water quality retrofits, often are not typically associated with water quality benefits.

4.8.2. Benefits of Adding Biochar

4.8.2.1. Verifiable Benefits

Biochar amendments to dry ponds are like those benefits seen with soil amendments and filter strips and embankments where stormwater function depends on in-situ soil structure. Biochar amendment to compacted or clayey soils can increase infiltration which supports runoff volume reduction. Biochar amendment to sandy soils can slightly decrease infiltration rate but increase water retention by increasing porosity and stable aggregates within bioretention media. This increased water retention also supports plant growth and survivability in harsh urban environments or during drought conditions.

4.8.2.2. Limited or No Benefits

Although pollutant reduction can be achieved through runoff (i.e. load) reduction and filtration, retrofitted dry ponds continue to exhibit the potential for sediment and associated pollutant resuspension due to low flow outlet proximity to the ground, and will likely not achieve improved pollutant removal credit by agencies without further modification, such as continuous monitoring and adaptive control (CMAC).

4.8.3. Biochar Application

Dry ponds are good candidates for biochar amendment when they exhibit signs of soil compaction or clogging, evident by bare soil patches during dry periods and standing water long after storms. Biochar amendment should be used to increase infiltration rates, reduce compaction, and transition a soil texture closer to a sandy loam which is the preferred soil texture for dry ponds (Figure 4.8.2).

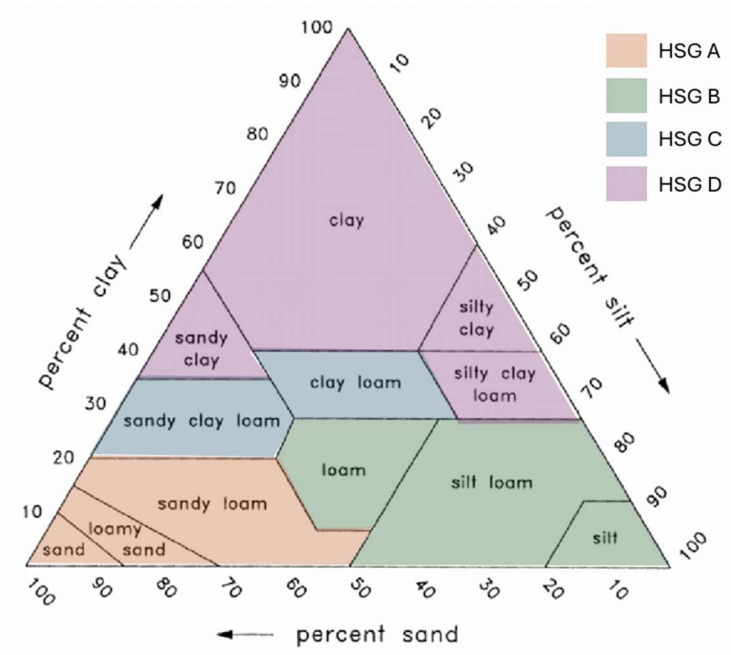


Figure 4.8.2. Soil texture and hydraulic soil group based on particles size distribution

4.8.3.1. Determine In-Situ Soil Texture

The first step in determining the best biochar application rate is to characterize in-situ soil conditions at the depth of proposed modification (usually 12 inches for dry ponds). Dry ponds soils are likely disturbed and not accurately represented on local or regional soil surveys. Therefore, field testing of identified biochar application areas is preferable to soil surveys. Field and lab soil tests may include:

- Particle Size Distribution (PSD) – Sand, silt, and clay composition (recommended).
- Organic Matter Content (% SOM) – Affects soil structure and moisture retention (optional).
- Bulk Density (g/cm^3) – Indicates compaction and aeration (required).
- Soil compaction or penetration testing (recommended)
- Saturated Hydraulic Conductivity (Ksat, in/hr) – Measures water movement through the soil in field (optional, good for quantifying improvements).
- Infiltration Rate (in/hr) – Determines surface water absorption capacity in field (optional, good for quantifying improvements).

4.8.3.2. Determine Biochar Application Rate

The recommended biochar application rates for dry pond retrofits are based on the site's existing soils and are between 5% and 10% by volume. Sites with USDA Hydraulic Soil Group (HSG) Class A, B and C soils are suitable for biochar soil amendments. However, biochar soil amendments for dry pond retrofits are most compatible with dry, sandy, well-drained soils, and/or soils with low nutrient content. HSG D soils can benefit from certain biochar applications, namely biochar material with low porosity to prevent enhanced retention, but there may be more suitable amendments for achieving similar results (i.e. sand, bioretention media). It is recommended that stormwater practitioners interested in improving stormwater performance within in-situ HSG D soils consult a local soil expert.

Biochar should not be applied at a greater rate than 10% by volume to prevent potential erosion due to concentrated flow through the BMP. If biochar at the recommended application rate is cost-prohibitive, adding less biochar will still benefit the existing soil conditions, but to a lesser extent.

4.8.3.3. Application Method

Amended areas should be cleared, subsoiled, tilled, amended, and stabilized within the same working day. Therefore, the size of the biochar soil amendment area should be carefully considered to ensure the work can be completed in a single working day. If the area is too large to complete in a single day, it can be broken into smaller areas to complete the amendment over the course of multiple days.

Existing vegetation should be removed, including turf, perennials, and overburden. Tillage for amendments should work outside of existing tree root zones. Subsoil – fracture the compacted soil found at depths 14 inches or deeper – to avoid adversely disturbing the topsoil. A Yeomans or similar plow is recommended for subsoiling the treatment area to the desired depth. Then a

rotary power harrow or other similar equipment for vertical tillage (do not use horizontal tillage) is used to till the freshly subsoiled soil to a depth of 12 inches. Biochar should then be applied and tilled into the soil in three passes, with approximately 1/3 of the total desired biochar amendment applied during each pass. Once all the required biochar has been thoroughly tilled into the in-situ soils, any tracks should be raked out, and any high areas should be leveled to minimize potential for concentrated runoff or ponding water. The disturbed area should be properly seeded with an appropriate permanent seed mix and stabilized with matting and check dam structures until vegetation is established.

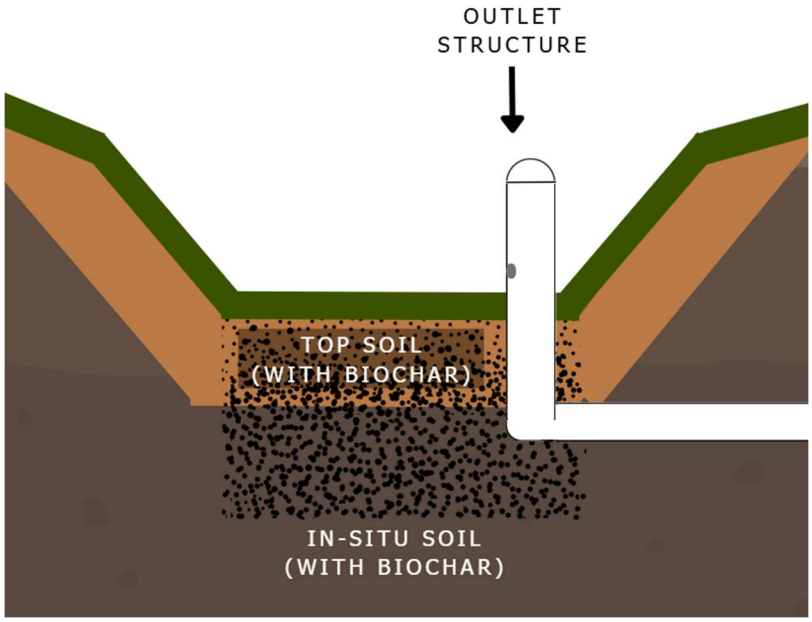


Figure 4.8.3 Graphic of biochar amendment to dry pond where the biochar amendment is incorporated into the in-situ soil.

4.8.4. Potential Concerns or Considerations

Major concerns from stormwater experts and installation crews when incorporating biochar into dry ponds are the challenges associated with moving and placing the biochar material. Biochar is light and can be easily scattered by the wind if not sufficiently moistened. Care should be taken by installers to ensure proper personal protective equipment (PPE) is used during biochar handling. On the other hand, like most media, biochar can become clumpy and difficult to apply uniformly if too wet. Striking the best moisture conditions for safety and ease of installation will come with experience as contractors work with biochar more frequently. To balance these concerns, a target moisture content of 25%-35% is recommended. Due to biochar's ability to conduct current, in-field testing using a probe will require calibration, but ultimately may help contractors maintain biochar material workability.

A biochar-amended soil will have a decreased bulk density and be susceptible to erosion in areas of concentrated flow, such as the base of the dry pond. Care should be taken to properly protect amended soils using erosion matting and check dams or wattles to prevent soil erosion and allow for vegetation establishment.

4.8.5. Specifications for Incorporating Biochar into Dry Pond

The following specifications templates are provided to guide material selection, storage and handling, and installation of biochar into dry detention ponds:

- **Biochar Material and Quality Control**
- **Biochar Material Handling**
- **Biochar Material Stockpiling**
- **Biochar-Amended Dry Detention Ponds**

5 Biochar Sourcing, Delivery, and Installation

5.1. Biochar Sourcing and Regulation

Biochar sourcing has transitioned from a fragmented market of small-scale producers to a more structured industry led by large-scale manufacturers capable of providing consistent product volumes. The reliable availability of material is now supported by the ABI, which maintains a registry of producers to facilitate identifying appropriate sources for specific applications, such as stormwater management.

Historically, the International Biochar Initiative (IBI) provided a voluntary certification program for standardizing product characterization, though this was discontinued as of 2024. Current sourcing and characterization efforts in North America are shifting toward ABI and ANSI standards. As of November 2025, it is recommended that biochar producers and end-users ensure that all material samples are analyzed according to the new ANSI S668: Methods for the Analysis and Testing of Biochar (LINK), which provides standardized testing protocols for properties such as carbon content, moisture, pH, and particle size distribution (PSD). These standards ensure compliance with North American regulations and laboratory capabilities.

5.2. Delivery Methods and Logistics

The method of biochar delivery is primarily determined by the transport distance, the intended application, and the scale of the project.

- Super Sacks (Bulk Bags): These are the preferred delivery method for long-distance hauls and are commonly utilized for stormwater applications like bioretention. They are typically delivered via box trucks or flatbed trailers and require heavy equipment, such as a forklift or crane, for unloading. Super sacks can be double stacked, with an average van trailer holding 48 to 52 bags.
- Bulk Shipments**:** This delivery method is typically reserved for shorter distances. Common vehicles include dump trucks or walking-floor trailers. For bulk loads, a maximum truck capacity is generally between 10 to 14 tons, though volume often limits the load before weight does, with biochar density frequently averaging 400 lb/cubic yard.
- Logistical Considerations**:** Moisture content significantly impacts transportation decisions; wetter biochar increases the total weight and associated shipping costs. For bulk shipping, a moisture content (MC) of 30% to 40% is often maintained to enhance safety and suppress dust during transit.

5.3. Biochar Storage and Handling

Proper storage and handling of biochar is critical to maintaining the biochar materials' stability and quality as well as mitigating any safety risks to personnel when receiving and storing shipments of biochar materials.

- Facility Design: Biochar should be stored in dry, well-ventilated areas to prevent the growth of mold or bacteria caused by moisture absorption. Large-scale storage may require non-combustible construction materials, such as steel-reinforced concrete, and firebreaks between storage units.

- Stockpiling: Bulk material stored in open piles should maintain a moisture content of approximately 50% to mitigate fire hazards. Piles deeper than 3 meters increase the risk of oxidative heating and spontaneous combustion.
- Super Sack Storage: These should be stored on pallets and covered to protect the material from UV degradation and saturation from rain events.

5.4. Biochar Implementation Challenges

The unique physicochemical properties of biochar introduce several potential operational challenges that can impact the sourcing, delivery, storage, and installation of biochar.

- Autoignition and Spontaneous Combustion: Biochar is a reactive material that can ignite at temperatures between 300°C and 400°C, depending on feedstock and pyrolysis conditions. Avoid storage environments with relative humidity below 60% to prevent moisture absorption, which can trigger microbial activity and spontaneous heating.
- Dust and Explosion Risks: Fine biochar particles (<2 mm) can form explosive dust clouds at concentrations exceeding 30 g/m³. Dust also poses a significant respiratory risk to personnel when biochar is significantly agitated and/or during transport.
- Inconsistent Product Quality: Variations in feedstock and pyrolysis conditions can alter the stability and performance of the biochar in stormwater BMPs. It is vital for both producers, soil vendors, and installers to test that biochar meets the appropriate material standards before implementation.
- Material Contamination: Biochar is highly hygroscopic and must be segregated from fertilizers or chemicals that could induce exothermic reactions or alter its stability.

5.5. Risk Mitigation and Specification Guidance

To mitigate risks to personnel and prevent the failure of stormwater BMPs, the following guidance and specification language may be integrated into specific stormwater BMP project guidance. Much of the following suggested specification language is included in the Section 6 – Specification Templates (See templates for *Biochar Handling*, *Biochar Materials & QC*, and *Biochar Stockpile*).

- **Personnel Safety & PPE:** Personnel handling biochar must wear appropriate Personal Protective Equipment (PPE), such as gloves, eye shields, and respirators to prevent skin contact and inhalation of fine particulates.
- **Quenching Specifications:** Material should be quenched in water post-pyrolysis to reduce flammability and dust formation. Biochar shall be rotated while adding water to ensure even saturation and prevent localized hotspots.
- **Fire Mitigation Specifications:** It is recommended that biochar be stored in dry storage for a minimum of 4 days prior to transport to ensure no spontaneous combustion occurs.
- **Transport Safety:** Avoid excessive agitation during transport to minimize dust generation, which can pose respiratory risks and fire hazards Bulk loads shall be stabilized using anti-static liners and load-securing nets or straps to prevent shifting and particulate release.

- **Monitoring:** Large-scale storage facilities should incorporate temperature monitoring systems with automated alerts to monitor temperature within stockpiles. Continuous infrared scans are recommended to detect subsurface hotspots in large piles.
- **BMP Integrity:** To ensure consistency in stormwater BMPs, specifications should require biochar testing per recommendation provided in Table A and Table B in Appendix A and Appendix B of this document.

6 SPECIFICATION TEMPLATES

Specification templates developed for the use of biochar in stormwater management BMPs related to this guidebook are available for download and review as a part of this guidebook. Specification templates can be found for the following topics:

- Biochar Material and Quality Control (LINK)
- Biochar Handling (LINK)
- Biochar Stockpile (LINK)
- Biochar & Compost Mixing for Soil Vendors (LINK)
- Biochar Soil Amendment (LINK)
- Biochar Dry Pond Amendment (LINK)
- Biochar Filter Strip & Embankment (LINK)
- Biochar – Vegetated Swale Amendments (LINK)
- Bioretention In-Situ Media with Biochar (LINK)
- Biochar-Enhanced Bioretention Media (BEBM) (LINK)
- Biochar for Urban Tree Plantings (LINK)

7 REFERENCES

Center for Watershed Protection & American Biochar Institute. (2026). Biochar in Stormwater BMPs Technical Basis Report. pp. 34.

APPENDICES

Appendix A – Table A. Recommended Parameters and Testing Responsibility for Biochar in Stormwater

Table A. Recommended parameters and testing responsibility for biochar in stormwater BMPs

Parameter	Laboratory Test[1]	Responsibility			BMP Type			
		Producer	Vendor	Installer	Green roofs	High Flow Media Filter	Bioretention with underdrains	Bioretention w/out underdrains; filter strip, soil amendment, vegetated swales, dry pond retrofit
Moisture	**ISO 18134-3**	**X**	**X**	**X**	25-35%	25-35%	25-35%	25-35%
Organic Carbon	**ISO 16948 per modifications in ANSI S668**	**X**	**X**		>75%	>75%	>75%	>75%
$H:C_{org}$	**ISO 16948 with ratio calculated per ANSI S668**	**X**	**X**		<0.4	<0.4	<0.4	<0.6
Total Ash	**ISO 18122**	**X**	**X**		<10%	<10%	<10%	<10%
Volatile Organic Matter	**ISO 18123**	**X**	**X**		<25%	<25%	<25%	<25%
Total Nitrogen	**ISO 16948**	**X**	**X**		<1,000 mg/kg	<1,000 mg/kg	<1,000 mg/kg	<2,000 mg/kg
Total Phosphorus	**ISO 16967, per ANSI S668**	**X**	**X**		<1,000 mg/kg	<1,000 mg/kg	<1,000 mg/kg	<2,000 mg/kg
Copper (Cu)	**Extraction via EPA 3050B or 3051A; Analysis via EPA 6010 / 6020, Hg via EPA 7471B or 7473**	**X**	**X**		<250 ppm	<250 ppm	<250 ppm	<250 ppm
Cadmium (Cd)		**X**	**X**		<10 ppm	<10 ppm	<10 ppm	<10 ppm
Lead (Pb)		**X**	**X**		<150 ppm	<150 ppm	<150 ppm	<150 ppm
Mercury (Hg)		**X**	**X**		<8 ppm	<8 ppm	<8 ppm	<8 ppm
Nickel (Ni)		**X**	**X**		<210 ppm	<210 ppm	<210 ppm	<210 ppm
Selenium (Se)		**X**	**X**		<18 ppm	<18 ppm	<18 ppm	<18 ppm
Zinc (Zn)		**X**	**X**		<1,400 ppm	<1,400 ppm	<1,400 ppm	<1,400 ppm
Arsenic (As)		**X**	**X**		<5 ppm	<5 ppm	<5 ppm	<5 ppm
pH	**ISO 10390**	**X**	**X**		<10	<11	<11	<11
Liming	**ISO 20978**	**X**	**X**		<5%	<10%	<10%	<10%

Parameter	Laboratory Test[1]	Responsibility			BMP Type			
		Producer	Vendor	Installer	Green roofs	High Flow Media Filter	Bioretention with underdrains	Bioretention w/out underdrains; filter strip, soil amendment, vegetated swales, dry pond retrofit
Electrical Conductivity	**ISO 11265 per modifications in ANSI S668**	**X**	**X**		<2 dS/m	<4 dS/m[2]	<2 dS/m	<2 dS/m
Particle Size Distribution (PSD)	**Progressive dry screening per ANSI S668**	**X**	**X**		#6 - 100% Passing; #18 <5% passing	#6 - 100% Passing; #30 <5% passing	#6 - 100% Passing; #40 <5% passing	#6 - 100% Passing; #100 <5% passing
Bulk Density	**ISO 17828**	**X**	**X**		6 to 12 lbs per cubic foot	6 to 12 lbs per cubic foot	6 to 12 lbs per cubic foot	6 to 12 lbs per cubic foot
PAHs	**Extraction via EPA 3550C or EPA 3540C w/ 100% toluene as extracting solvent; Analysis via EPA 8270E; Report total + by compound**	**X**	**X**		<300 ppm	<300 ppm	<300 ppm	<300 ppm

[1] Testing method as specified in ANSI S668 - Methods for Measurement and Testing of Biochar. All sample preparation shall follow this document unless otherwise noted.
[2] For biochar >2 dS/m, consider pre-washing biochar and avoid planting salt-sensitive plant species.

Appendix B – Table B. Biochar Characteristics and Additional Parameters for Stormwater Management Goals

Table B. Biochar characteristics and additional parameters thresholds for removal of target pollutants in stormwater BMPs.

Parameter	Laboratory Test[1]	Acceptable Range							
		Responsibility			Target Pollutant				
		Producer	Vendor	Installer	Heavy Metals	Nitrogen	Phosphorus	Microbes	Trace Organic Contaminants
$H:C_{org}$	**ISO 16948 with ratio calculated per ANSI S668**	**X**	**X**		<0.3	<0.6	<0.6	<0.3	<0.3
Nitrogen Leaching	**Pre-leach biochar by washing 10x pore volume of liquid through sample; EPA Method 1312, EPA Method 353.2, EPA Method 350.1**	**X**	**X**			<0.15 mg/L			
Phosphorus Leaching	**Pre-leach biochar by washing 10x pore volume of liquid through sample; EPA Method 1312, EPA Method 353.2, EPA Method 350.1**	**X**	**X**				<0.15 mg/L		
Specific Surface Area (SSA)	**ISO 9277 following guidance per ANSI S668**	**X**	**X**		>300 m2/g			>300 m2/g	>300 m2/g

[1] Testing method as specified in ANSI S668 - Methods for Measurement and Testing of Biochar. All sample preparation shall follow this document unless otherwise noted.